# BMI and documented hypercapnia in acute care

**Expanded scientific report for investigator review**

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**Protected research draft. Investigator interpretation and scientific approval remain pending.**

Saved run: curve-analyze-20261024T053607-55109. Analysis: bmi_hypercapnia_rfs_curve_audit_v7.

Source: identity_bound_source_declarations_passed; analysis completed: TRUE. Review states in the immutable bundle are recorded at estimation time; later signed review receipts are separate.

## 1 Introduction

We describe BMI–hypercapnia relationships within a selected acute-care cohort while exposing how historical source-entry pathways shape the observed population. These pathways record representation in reason-for-suspicion (RFS) source files; they do not necessarily identify every clinical criterion present at an encounter. Separating entry routes can change the observed association itself and does not eliminate outcome-dependent selection.

# 2 Methods

## 2.1 Source, eligibility and fixed index

The authenticated original FULL_DATA encounter list defines candidate visits. Adults had a 2022 emergency or inpatient encounter and available valid BMI ≥30 kg/m². Age uses the inherited birth-year proxy. Each patient contributes the first qualifying encounter before gas availability and adjustment completeness. Earlier unresolved eligibility, conflicting evidence and earliest-date ties remain unresolved. Later encounters never replace an obesity-only index.

BMI follows the approved encounter/prior reported-value hierarchy, followed by calculation from observed height and weight, with inherited rounding and validity rules. One-unit diagnostic bins around BMI 40 and 50 describe measurement and selection; they are not a regression-discontinuity design. No statistical BMI imputation is performed. Available BMI is not necessarily contemporaneous with the outcome. The audit reports measurement methods, lags, unknown chronology and consequences of the existing rules. Signed lags equal encounter-start date minus measurement date: positive means prior, zero means the same calendar day, and negative means later during the encounter. Component gaps are absolute differences between height and weight dates.

## 2.2 Historical source-entry populations

A confirmed positive ABG, VBG, RESPFAIL, VENTSUPPORT or PREDISPOSITION source flag establishes the primary population, including encounters also represented through OBESITY. Obesity-only membership requires OBESITY and authenticated zeros in all five other families. Obesity-only indexes are excluded from scientific comparisons: removing the gas-related entry pathways selects against the evidence needed to assess the physiological outcomes. Their counts remain in the flow for reconciliation; source discrepancies and descriptive profiles are confined to the audit. Unknown flags remain unknown. All-six-zero discrepancies require source investigation. RFS metadata are applied after the common index is selected.

The combined indexed cohort is a reconciliation reference. Non-OBESITY-only, overlapping and unknown-OBESITY/confirmed-non-OBESITY subgroups are descriptive. BMI ≥40 remains a descriptive stratum and is not equated with obesity-route membership. An overlapping all-OBESITY supporting analysis is confined to the audit.

## 2.3 Outcomes and adjustment

Arterial hypercapnia requires ordinary authenticated ABG $PaCO_2$ ≥45 mm Hg. Combined gas accepts ABG $PaCO_2$ ≥45 or VBG $PvCO_2$ ≥50 mm Hg; both specimens are not required. No pH restriction is applied to the established primary outcomes. Strict validity limits are 5<arterial<200 and 10<venous<250 mm Hg, with reviewed units. Available-evidence negatives do not establish that an unmeasured specimen is normal. Unusable or unavailable outcomes remain unknown.

The ICD composite comprises J96.02, J96.12, J96.22, J96.92 and E66.2, with captured absence negative only under complete diagnosis capture. E66.2 introduces obesity-related ascertainment. Index-calendar-day gas sensitivities use recorded-result date equal to encounter-start date, retaining the index. This export does not authenticate collection times.

Adjustment retains linear age and the inherited sex, combined race/ethnicity and geographic-region coding. Completeness means nonmissing values for inherited analysis variables, including retained Unknown categories. The White reference category includes non-Hispanic or unknown ethnicity; inherited race-category precedence is unchanged. Raw race × ethnicity cross-tabs are shown only when authenticated.

### 2.4 Models, probabilities and uncertainty

Primary models use type-7 BMI quantiles .05/.35/.65/.95 from the primary adjustment-complete population: two interior knots, two boundaries and three natural-spline basis columns. The basis remains fixed across primary outcomes, timing, matched analyses and predictions. The primary current-measurement sensitivity reuses this basis. The overlapping all-OBESITY supporting analysis uses its own basis and is reported in the audit. Obesity-only and obesity-only current-measurement models are not registered. Constant outcomes, duplicate knots, separation, rank and convergence failures in planned models remain explicit without estimator substitution.

Gas curves standardize over each model's outcome-evaluable population; ICD uses known capture within group. Observation curves describe usable gas-evidence availability. Complete diagnosis capture is reported as a source fact, not as ICD positivity or a constant availability curve. No physiological outcomes are projected to untested patients. Fixed BMI 45 versus 40 contrasts require training-range support. Local counts and marginal covariate coverage expose sparsity even inside the global BMI range; these checks do not establish joint covariate support.

Pointwise 95% intervals use full coefficient covariance and condition on the empirical reference and fixed spline basis. They omit uncertainty from selection, measurement error and unmeasured confounding. Apparent calibration is in-sample; fitted/observed mean agreement is not independent validation.

### 2.5 Secondary scenario and supporting checks

The named secondary scenario subtracts exactly one BMI unit when observed BMI >30; BMI=30 is unchanged. There is no floor or reselection. Paired predictions hold index, RFS membership and other covariates fixed. Reported quantities are observed and shifted mean probabilities, their signed difference and PIF=1−p_shifted/p_observed. Signed original-BMI-band contributions and their full covariance reproduce the total difference and variance. Marginal curve-density integration is not used.

Same-sample linear-BMI checks and matched ICD/gas analyses remain registered. A limited current-measurement sensitivity uses the already-selected BMI when reported at the index or calculated from height and weight both authenticated to that encounter. Full-cohort MAR/MNAR projections and IPW are excluded. The report is rendered from saved aggregates without clinical reads or refitting.

## 3 Results

### 3.1 Primary cohort and usable evidence

Table 1: Fixed-index RFS partition

| population id | n | complete | incomplete | denominator | status |
|---|---|---|---|---|---|
| primary | 176922 | 176922 | 0 | 223514 | assessed |
| obesity only | 46592 | 46592 | 0 | 223514 | assessed |
| unresolved | 0 | 0 | 0 | 223514 | assessed |

## Source-entry partition at the fixed index

Mutually exclusive patient groups; overlap remains in primary

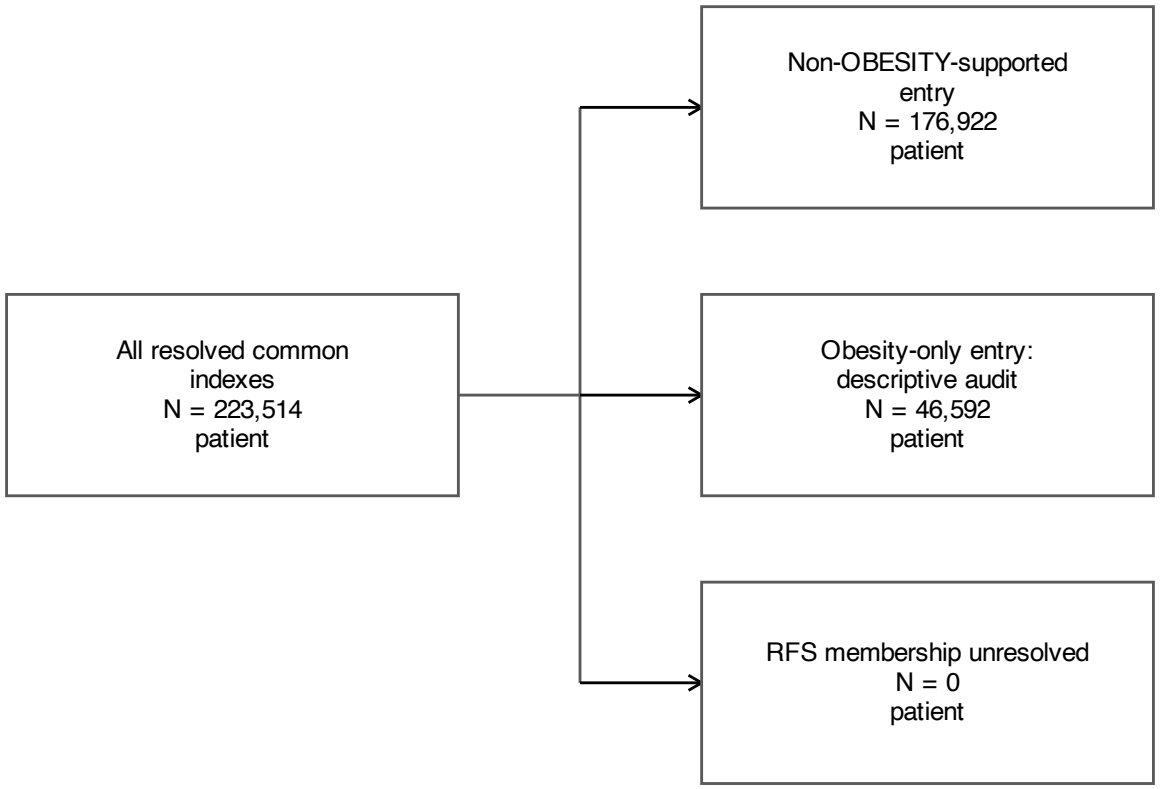



Figure 1: Fixed common indexes partition into primary non-OBESITY-supported source entry, obesity-only source entry and unresolved RFS. Overlapping OBESITY/non-OBESITY membership remains primary. Counts and units derive from the ledger; all-OBESITY supporting models overlap and are not a fourth partition branch.

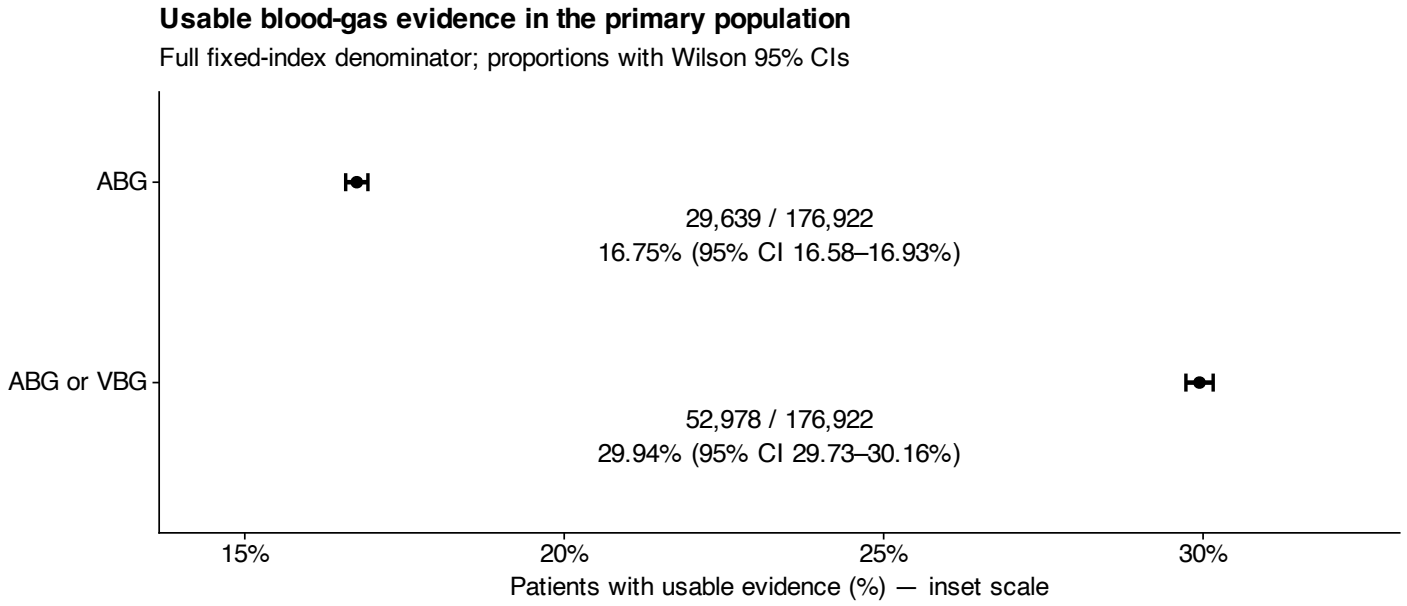



Figure 2: Encounter-wide usable gas evidence among all primary fixed indexes, before adjustment completeness. Labels give evaluable/full N, percentage and 95% Wilson CI; capped intervals use their exact endpoints on a labeled inset scale. ABG: arterial blood gas; VBG: venous blood gas. The combined row accepts either usable specimen; the two rows overlap and are not summed. These percentages describe evidence availability, not hypercapnia positivity. Primary includes non-OBESITY pathways and their OBESITY overlaps. Complete diagnosis capture, when present, is a source fact rather than ICD positivity and is reported separately. Obesity-only counts and residual-gas discrepancies are confined to the cohort flow and audit; they do not define a scientific comparator or weighting analysis.

Diagnosis capture was known for 176,922 of 176,922 primary indexes. This describes source coverage, not the presence of an ICD hypercapnia code; observed ICD positivity is reported below. The complete source-entry reconciliation and residual-gas audit are in the companion audit report.

### 3.2 Source-entry population: primary

grain: patient; status: assessed; analysis population id: primary.

Table 2: Population and evidence accounting

| population id | label | n records | n encounters | n patients | n |
|---|---|---|---|---|---|
| Q | Resolved first indexes (Q) | 176922 | 176922 | 176922 | 176922 |
| missing adjustment | missing adjustment | 0 | 0 | 0 | 0 |
| C | Complete adjustment (C) | 176922 | 176922 | 176922 | 176922 |
| C icd | Known diagnosis capture (C icd) | 176922 | 176922 | 176922 | 176922 |
| C abg encounter | C abg encounter | 29639 | 29639 | 29639 | 29639 |
| C any gas encounter | C any gas encounter | 52978 | 52978 | 52978 | 52978 |

**variable_id=age_at_encounter**

n missing: 0; status: assessed; minimum: 18; q95: 82; q99: 87; n invalid: 0; analysis population id: primary.

Table 3: Participant profiles — panel 1/2

| population id | n | n observed | n distinct | mean | sd | q01 | q05 |
|---|---|---|---|---|---|---|---|
| Q | 176922 | 176922 | 77 | 57.47882 | 16.77562 | 21 | 27 |
| C | 176922 | 176922 | 77 | 57.47882 | 16.77562 | 21 | 27 |
| C abg encounter | 29639 | 29639 | 75 | 61.57765 | 14.93598 | 23 | 32 |
| C any gas encounter | 52978 | 52978 | 76 | 59.75945 | 15.68863 | 22 | 30 |

Table 4: Participant profiles — panel 2/2

| population id | q25 | median | q75 | maximum |
|---|---|---|---|---|
| Q | 45 | 60 | 71 | 96 |
| C | 45 | 60 | 71 | 96 |
| C abg encounter | 53 | 64 | 73 | 92 |
| C any gas encounter | 50 | 62 | 72 | 96 |

**variable_id=bmi**

n missing: 0; status: assessed; minimum: 30; q01: 30; q95: 49.7000007629395; maximum: 89.8000030517578; n invalid: 0; analysis population id: primary.

Table 5: Participant profiles — panel 1/2

| population id | n | n observed | n distinct | mean | sd | q05 | q25 |
|---|---|---|---|---|---|---|---|
| Q | 176922 | 176922 | 488 | 37.903 | 6.699107 | 30.4 | 32.6 |
| C | 176922 | 176922 | 488 | 37.903 | 6.699107 | 30.4 | 32.6 |
| C abg encounter | 29639 | 29639 | 436 | 37.01368 | 6.687356 | 30.3 | 32.1 |
| C any gas encounter | 52978 | 52978 | 458 | 37.12698 | 6.652777 | 30.3 | 32.2 |

Table 6: Participant profiles — panel 2/2

| population id | median | q75 | q99 |
|---|---|---|---|
| Q | 36.3 | 41.6 | 60.5 |
| C | 36.3 | 41.6 | 60.5 |
| C abg encounter | 35 | 39.9 | 61.3 |
| C any gas encounter | 35.2 | 40.1 | 60.8 |

**variable_id=sex**

Unavailable throughout this group: mean, sd, minimum, q01, q05, q25, median, q75, q95, q99, maximum, n_invalid.

Table 7: Participant profiles

| population id | n | n observed | n missing | n distinct | status | analysis population id |
|---|---|---|---|---|---|---|
| Q | 176922 | 176922 | 0 | 2 | assessed | primary |
| C | 176922 | 176922 | 0 | 2 | assessed | primary |
| C abg encounter | 29639 | 29639 | 0 | 2 | assessed | primary |
| C any gas encounter | 52978 | 52978 | 0 | 2 | assessed | primary |

**variable_id=race_ethnicity**

Unavailable throughout this group: mean, sd, minimum, q01, q05, q25, median, q75, q95, q99, maximum, n_invalid.

Table 8: Participant profiles

| population id | n | n observed | n missing | n distinct | status | analysis population id |
|---|---|---|---|---|---|---|
| Q | 176922 | 176922 | 0 | 7 | assessed | primary |
| C | 176922 | 176922 | 0 | 7 | assessed | primary |
| C abg encounter | 29639 | 29639 | 0 | 7 | assessed | primary |
| C any gas encounter | 52978 | 52978 | 0 | 7 | assessed | primary |

**variable_id=location**

Unavailable throughout this group: mean, sd, minimum, q01, q05, q25, median, q75, q95, q99, maximum, n_invalid.

Table 9: Participant profiles

| population id | n | n observed | n missing | n distinct | status | analysis population id |
|---|---|---|---|---|---|---|
| Q | 176922 | 176922 | 0 | 4 | assessed | primary |
| C | 176922 | 176922 | 0 | 4 | assessed | primary |
| C abg encounter | 29639 | 29639 | 0 | 4 | assessed | primary |
| C any gas encounter | 52978 | 52978 | 0 | 4 | assessed | primary |

Arterial $PCO_2$ ≥45 mm Hg: 15,541 documented-positive outcomes among 29,639 model patients; observed positivity 52.43% (95% CI 51.87–53%) (Wilson interval). Model status: successful.

Arterial ≥45 or venous ≥50 mm Hg: 23,621 documented-positive outcomes among 52,978 model patients; observed positivity 44.59% (95% CI 44.16–45.01%) (Wilson interval). Model status: successful.

Documented ICD composite: 8,634 documented-positive outcomes among 176,922 model patients; observed positivity 4.88% (95% CI 4.781–4.981%) (Wilson interval). Model status: successful.

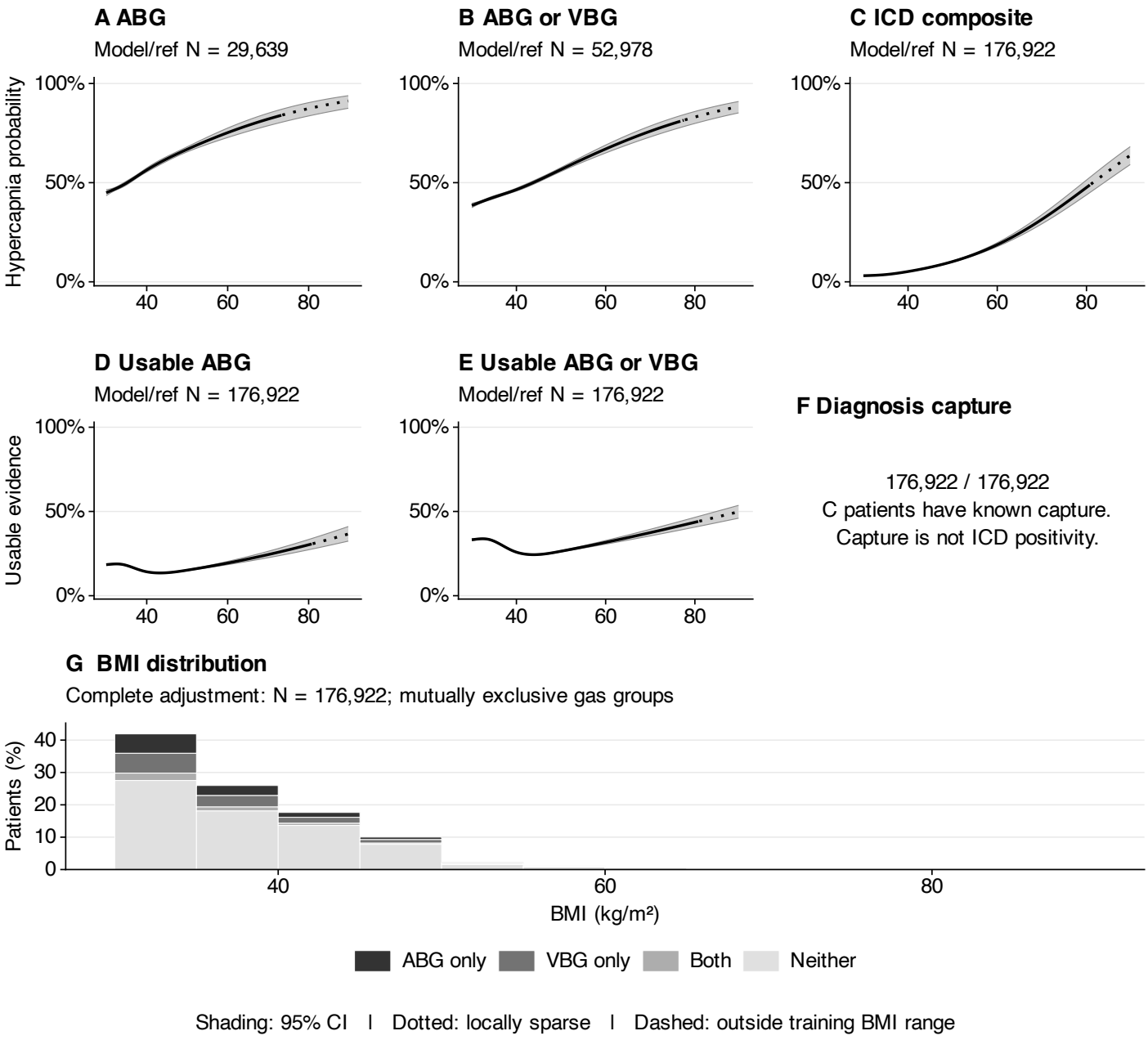



Figure 3: Source-entry population: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F1. BMI–hypercapnia relationships at the first eligible adult 2022 acute-care encounter. ABG denotes $PaCO_2$ ≥45 mm Hg; ABG or VBG denotes $PaCO_2$ ≥45 or $PvCO_2$ ≥50 mm Hg. ICD denotes the five-code composite, including E66.2. Gas curves standardize over outcome-evaluable patients; ICD uses known diagnosis capture. Model/ref N is the common model and reference count. Availability and BMI frequency use all adjustment-complete patients (C); the four frequency groups are mutually exclusive. Adjustment: age, sex, race/ethnicity and region with a fixed natural BMI spline. Shading and fine boundaries show pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. Open availability points use Wilson intervals when no model is estimable. Known capture is distinct from ICD positivity. ABG/VBG: arterial/venous blood gas. Conditional model uncertainty excludes selection and measurement assumptions; no population transport or causal effect is inferred.

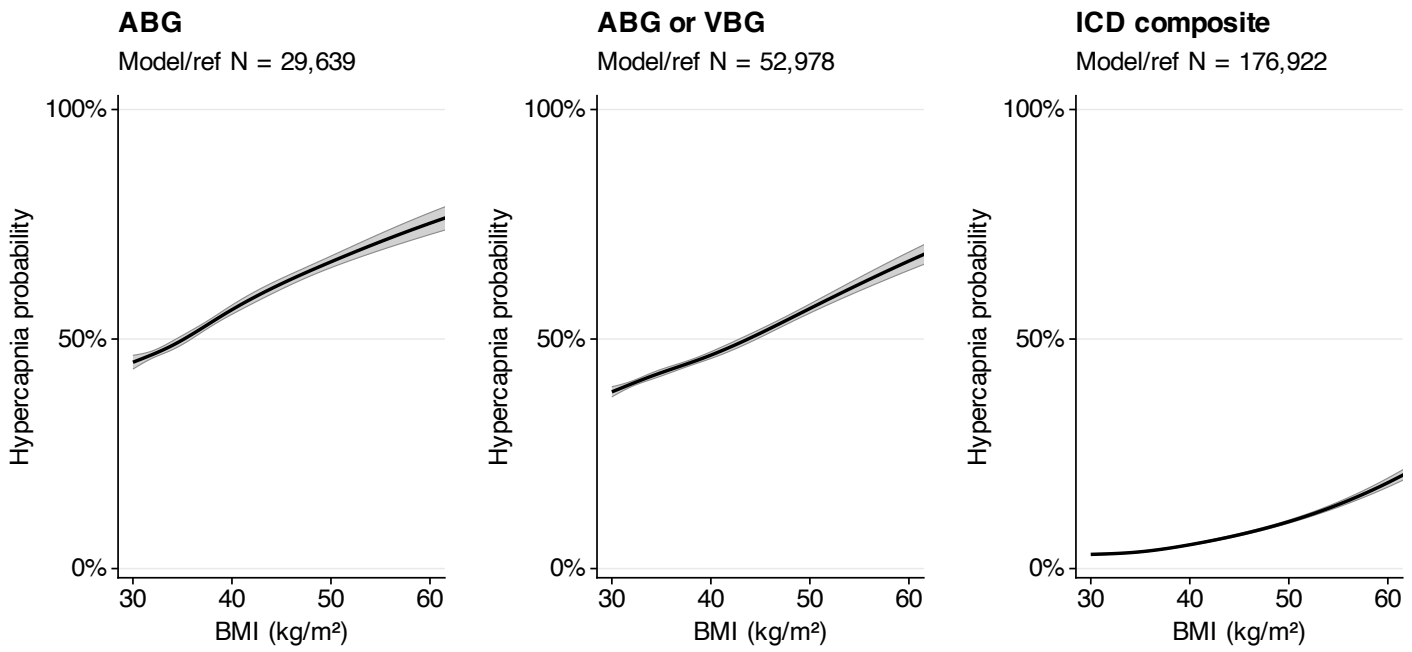



Figure 4: Source-entry population: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F1. BMI–hypercapnia relationships at the first eligible adult 2022 acute-care encounter. ABG denotes $PaCO_2$ ≥45 mm Hg; ABG or VBG denotes $PaCO_2$ ≥45 or $PvCO_2$ ≥50 mm Hg. ICD denotes the five-code composite, including E66.2. Gas curves standardize over outcome-evaluable patients; ICD uses known diagnosis capture. Model/ref N is the common model and reference count. Availability and BMI frequency use all adjustment-complete patients (C); the four frequency groups are mutually exclusive. Adjustment: age, sex, race/ethnicity and region with a fixed natural BMI spline. Shading and fine boundaries show pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. Open availability points use Wilson intervals when no model is estimable. Known capture is distinct from ICD positivity. ABG/VBG: arterial/venous blood gas. Conditional model uncertainty excludes selection and measurement assumptions; no population transport or causal effect is inferred. This labeled detail view limits the displayed BMI axis to 30–60; no observations were removed from estimation.

bmi low: 40; bmi high: 45; status: estimated; reason: ; analysis population id: primary; population id: primary; ascertainment window: encounter; model form: spline.

Table 10: Supported BMI 45 versus 40 contrasts — panel 1/3

| model id | reference id | n reference | n model | probability low bmi | probability high bmi | probability low bmi low | probability low bmi high |
|---|---|---|---|---|---|---|---|
| abg/encounter/spline | C abg encounter | 29639 | 29639 | 0.563898 | 0.6204125 | 0.5538255 | 0.5739181 |
| any gas/encounter/spline | C any gas encounter | 52978 | 52978 | 0.4649322 | 0.5127818 | 0.4575954 | 0.4722842 |
| icd/encounter/spline | C icd | 176922 | 176922 | 0.05191386 | 0.07372897 | 0.05038018 | 0.0534916 |

Table 11: Supported BMI 45 versus 40 contrasts — panel 2/3

| model id | probability high bmi low | probability high bmi high | difference per1000 | conf low per1000 | conf high per1000 | prevalence ratio | ratio low |
|---|---|---|---|---|---|---|---|
| abg/encounter/spline | 0.6086704 | 0.6320152 | 56.51448 | 50.37713 | 62.65183 | 1.100221 | 1.089214 |
| any gas/encounter/spline | 0.5038924 | 0.521663 | 47.84958 | 43.14841 | 52.55074 | 1.102917 | 1.092791 |
| icd/encounter/spline | 0.07118096 | 0.07636069 | 21.81511 | 20.24304 | 23.38718 | 1.420217 | 1.392341 |

Table 12: Supported BMI 45 versus 40 contrasts — panel 3/3

| model id | ratio high | reference population id | outcome id |
|---|---|---|---|
| abg/encounter/spline | 1.111339 | C abg encounter | abg |
| any gas/encounter/spline | 1.113138 | C any gas encounter | any gas |
| icd/encounter/spline | 1.448652 | C icd | icd |

Arterial $PCO_2$ ≥45 mm Hg: standardized probability at BMI 40 was 56.39% and at BMI 45 was 62.04%. The difference was 5.651 (5.038, 6.265) percentage points (95% CI), within C_abg_encounter. This is a fixed-BMI contrast, not the one-unit distribution-shift PIF.

Arterial ≥45 or venous ≥50 mm Hg: standardized probability at BMI 40 was 46.49% and at BMI 45 was 51.28%. The difference was 4.785 (4.315, 5.255) percentage points (95% CI), within C_any_gas_encounter. This is a fixed-BMI contrast, not the one-unit distribution-shift PIF.

Documented ICD composite: standardized probability at BMI 40 was 5.191% and at BMI 45 was 7.373%. The difference was 2.182 (2.024, 2.339) percentage points (95% CI), within C_icd. This is a fixed-BMI contrast, not the one-unit distribution-shift PIF.

quantile type: 7; status: resolved; reference id: C; basis: splines::ns; intercept FALSE; 3 columns; analysis population id: primary; reference population id: C.

Table 13: Saved group-specific numeric basis

| basis id | knot probability | knot | kind |
|---|---|---|---|
| 026b7a1829cd49840558b84e972374c0e1cb1f3a2f2304811582736b3d37f105 | 0.05 | 30.4 | boundary |
| 026b7a1829cd49840558b84e972374c0e1cb1f3a2f2304811582736b3d37f105 | 0.35 | 33.9 | interior |
| 026b7a1829cd49840558b84e972374c0e1cb1f3a2f2304811582736b3d37f105 | 0.65 | 39.3 | interior |

| basis id | knot probability | knot | kind |
|---|---|---|---|
| 026b7a1829cd49840558b84e972374c0e1cb1f3a2f2304811582736b3d37f105 | 0.95 | 49.7 | boundary |

Table 14: Encounter-wide spline fit status

| Field | Value |
|---|---|
| model id | abg/encounter/spline |
| family | primary |
| outcome id | abg |
| window | encounter |
| training population id | C abg encounter |
| field | abg |
| model form | spline |
| source status | qualified |
| formula text | .outcome ~ bmi ns1 + bmi ns2 + bmi ns3 + age at encounter + sex + race ethnicity + location |
| basis id | 026b7a1829cd49840558b84e972374c0e1cb1f3a2f2304811582736b3d37f105 |
| weight specification | none |
| covariance method | model-based binomial; empirical reference fixed |
| n eligible | 176922 |
| n excluded outcome | 147283 |
| n excluded adjustment | 0 |
| n model | 29639 |
| n patients | 29639 |
| n events | 15541 |
| n nonevents | 14098 |
| n parameters | 15 |
| model rank | 15 |
| status | successful |
| failure reason | |
| warnings | |
| fit runtime | 0.152 |
| analysis population id | primary |
| ascertainment window | encounter |
| population id | primary |
| source reason | unavailable |
| estimator | unavailable |
| separation detected | unavailable |

Table 15: Complete primary coefficients

| Term | Estimate | SE | 95% CI | P value | Interpretation |
|---|---|---|---|---|---|
| (Intercept) | -0.7443513 | 0.06549514 | -0.7444 (-0.8727, -0.616) | 6.247462e-30 | log-odds contrast |
| bmi ns1 | 0.4774925 | 0.04381378 | 0.4775 (0.3916, 0.5634) | 1.175264e-27 | basis coefficient; no standalone clinical OR |
| bmi ns2 | 0.8627809 | 0.06496352 | 0.8628 (0.7355, 0.9901) | 2.983564e-40 | basis coefficient; no standalone clinical OR |
| bmi ns3 | 0.8146604 | 0.03563716 | 0.8147 (0.7448, 0.8845) | 1.166056e-115 | basis coefficient; no standalone clinical OR |
| age at encounter | 0.00650236 | 8.20106e-04 | 0.006502 (0.004895, 0.00811) | 2.214828e-15 | log-odds contrast |
| sex1 | 0.3179027 | 0.02403412 | 0.3179 (0.2708, 0.365) | 6.117298e-40 | log-odds contrast |
| race ethnicity1 | -0.110524 | 0.03192594 | -0.1105 (-0.1731, -0.04795) | 5.364e-04 | log-odds contrast |
| race ethnicity2 | -0.1190858 | 0.04802947 | -0.1191 (-0.2132, -0.02495) | 0.0131592 | log-odds contrast |

| Term | Estimate | SE | 95% CI | P value | Interpretation |
|---|---|---|---|---|---|
| race ethnicity3 | -0.1713243 | 0.1225482 | -0.1713 (-0.4115, 0.06887) | 0.1621085 | log-odds contrast |
| race ethnicity4 | -0.139982 | 0.1188708 | -0.14 (-0.373, 0.093) | 0.2389568 | log-odds contrast |
| race ethnicity5 | -0.1841403 | 0.2568364 | -0.1841 (-0.6875, 0.3192) | 0.4734015 | log-odds contrast |
| race ethnicity6 | -0.02230447 | 0.0486884 | -0.0223 (-0.1177, 0.07312) | 0.6468759 | log-odds contrast |
| location1 | 0.1168835 | 0.03160167 | 0.1169 (0.05495, 0.1788) | 2.167504e-04 | log-odds contrast |
| location2 | 0.04044335 | 0.03783289 | 0.04044 (-0.03371, 0.1146) | 0.2850698 | log-odds contrast |
| location3 | -0.02878643 | 0.03477898 | -0.02879 (-0.09695, 0.03938) | 0.4078425 | log-odds contrast |
| sex=0 | unavailable | unavailable | Reference; not estimated | unavailable | reference level |
| race ethnicity=0 | unavailable | unavailable | Reference; not estimated | unavailable | reference level |
| location=0 | unavailable | unavailable | Reference; not estimated | unavailable | reference level |

model id: abg/encounter/spline; note: apparent in-sample diagnostic; no validation or causal claim; analysis population id: primary; population id: primary; outcome id: abg; ascertainment window: encounter; model form: spline.

Table 16: Apparent fit diagnostics

| metric | value |
|---|---|
| log likelihood | -20102.16 |
| null deviance | 41018.1 |
| residual deviance | 40204.33 |
| AIC | 40234.33 |
| Brier | 0.2426885 |
| log loss | 0.6782335 |
| iterations | 4 |
| residual df | 29624 |
| scaled condition number | 9.720362 |
| events per parameter | 1036.067 |
| nonevents per parameter | 939.8667 |

Table 17: Encounter-wide spline fit status

| Field | Value |
|---|---|
| model id | any gas/encounter/spline |
| family | primary |
| outcome id | any gas |
| window | encounter |
| training population id | C any gas encounter |
| field | any gas |
| model form | spline |
| source status | qualified |
| formula text | .outcome ~ bmi ns1 + bmi ns2 + bmi ns3 + age at encounter + sex + race ethnicity + location |
| basis id | 026b7a1829cd49840558b84e972374c0e1cb1f3a2f2304811582736b3d37f105 |
| weight specification | none |
| covariance method | model-based binomial; empirical reference fixed |
| n eligible | 176922 |
| n excluded outcome | 123944 |
| n excluded adjustment | 0 |
| n model | 52978 |
| n patients | 52978 |
| n events | 23621 |

| Field | Value |
|---|---|
| n nonevents | 29357 |
| n parameters | 15 |
| model rank | 15 |
| status | successful |
| failure reason | |
| warnings | |
| fit runtime | 0.235 |
| analysis population id | primary |
| ascertainment window | encounter |
| population id | primary |
| source reason | unavailable |
| estimator | unavailable |
| separation detected | unavailable |

Table 18: Complete primary coefficients

| Term | Estimate | SE | 95% CI | P value | Interpretation |
|---|---|---|---|---|---|
| (Intercept) | -1.303706 | 0.04848148 | -1.304 (-1.399, -1.209) | 2.812911e-159 | log-odds contrast |
| bmi ns1 | 0.2784644 | 0.03225256 | 0.2785 (0.2153, 0.3417) | 5.931061e-18 | basis coefficient; no standalone clinical OR |
| bmi ns2 | 0.6974113 | 0.04972655 | 0.6974 (0.5999, 0.7949) | 1.097228e-44 | basis coefficient; no standalone clinical OR |
| bmi ns3 | 0.6437564 | 0.02590718 | 0.6438 (0.593, 0.6945) | 2.679285e-136 | basis coefficient; no standalone clinical OR |
| age at encounter | 0.01352009 | 5.983956e-04 | 0.01352 (0.01235, 0.01469) | 4.975362e-113 | log-odds contrast |
| sex1 | 0.2718704 | 0.01806465 | 0.2719 (0.2365, 0.3073) | 3.459564e-51 | log-odds contrast |
| race ethnicity1 | -0.04276667 | 0.02336648 | -0.04277 (-0.08856, 0.003031) | 0.0672115 | log-odds contrast |
| race ethnicity2 | -0.08115003 | 0.03653968 | -0.08115 (-0.1528, -0.009534) | 0.02635946 | log-odds contrast |
| race ethnicity3 | -0.2250825 | 0.0963428 | -0.2251 (-0.4139, -0.03625) | 0.01947734 | log-odds contrast |
| race ethnicity4 | -0.4527456 | 0.08747207 | -0.4527 (-0.6242, -0.2813) | 2.268291e-07 | log-odds contrast |
| race ethnicity5 | -0.1230116 | 0.2138289 | -0.123 (-0.5421, 0.2961) | 0.5651018 | log-odds contrast |
| race ethnicity6 | -0.1907534 | 0.03253794 | -0.1908 (-0.2545, -0.127) | 4.559761e-09 | log-odds contrast |
| location1 | -0.05575314 | 0.02172421 | -0.05575 (-0.09833, -0.01317) | 0.01027584 | log-odds contrast |
| location2 | -0.1264609 | 0.02926886 | -0.1265 (-0.1838, -0.06909) | 1.555619e-05 | log-odds contrast |
| location3 | -0.1643927 | 0.02879889 | -0.1644 (-0.2208, -0.1079) | 1.141105e-08 | log-odds contrast |
| sex=0 | unavailable | unavailable | Reference; not estimated | unavailable | reference level |
| race ethnicity=0 | unavailable | unavailable | Reference; not estimated | unavailable | reference level |
| location=0 | unavailable | unavailable | Reference; not estimated | unavailable | reference level |

model id: any_gas/encounter/spline; note: apparent in-sample diagnostic; no validation or causal claim; analysis population id: primary; population id: primary; outcome id: any_gas; ascertainment window: encounter; model form: spline.

Table 19: Apparent fit diagnostics

| metric | value |
|---|---|
| log likelihood | -35618.53 |
| null deviance | 72820.84 |
| residual deviance | 71237.06 |
| AIC | 71267.06 |
| Brier | 0.2398443 |
| log loss | 0.6723268 |
| iterations | 4 |
| residual df | 52963 |
| scaled condition number | 9.58629 |
| events per parameter | 1574.733 |
| nonevents per parameter | 1957.133 |

Table 20: Encounter-wide spline fit status

| Field | Value |
|---|---|
| model id | icd/encounter/spline |
| family | primary |
| outcome id | icd |
| window | encounter |
| training population id | C icd |
| field | icd |
| model form | spline |
| source status | qualified |
| formula text | .outcome ~ bmi ns1 + bmi ns2 + bmi ns3 + age at encounter + sex + race ethnicity + location |
| basis id | 026b7a1829cd49840558b84e972374c0e1cb1f3a2f2304811582736b3d37f105 |
| weight specification | none |
| covariance method | model-based binomial; empirical reference fixed |
| n eligible | 176922 |
| n excluded outcome | 0 |
| n excluded adjustment | 0 |
| n model | 176922 |
| n patients | 176922 |
| n events | 8634 |
| n nonevents | 168288 |
| n parameters | 15 |
| model rank | 15 |
| status | successful |
| failure reason | |
| warnings | |
| fit runtime | 0.789 |
| analysis population id | primary |
| ascertainment window | encounter |
| population id | primary |
| source reason | unavailable |
| estimator | unavailable |
| separation detected | unavailable |

Table 21: Complete primary coefficients

| Term | Estimate | SE | 95% CI | P value | Interpretation |
|---|---|---|---|---|---|
| (Intercept) | -5.447039 | 0.06739039 | -5.447 (-5.579, -5.315) | 0 | log-odds contrast |
| bmi ns1 | 0.5966077 | 0.03621438 | 0.5966 (0.5256, 0.6676) | 5.610631e-61 | basis coefficient; no standalone clinical OR |
| bmi ns2 | 1.092816 | 0.07349178 | 1.093 (0.9488, 1.237) | 5.168485e-50 | basis coefficient; no standalone clinical OR |
| bmi ns3 | 1.214574 | 0.02970283 | 1.215 (1.156, 1.273) | 0 | basis coefficient; no standalone clinical OR |

| Term | Estimate | SE | 95% CI | P value | Interpretation |
|---|---|---|---|---|---|
| age at encounter | 0.03166113 | 7.927538e-04 | 0.03166 (0.03011, 0.03321) | 0 | log-odds contrast |
| sex1 | 0.1788576 | 0.02292163 | 0.1789 (0.1339, 0.2238) | 6.044934e-15 | log-odds contrast |
| race ethnicity1 | -0.1356647 | 0.03149877 | -0.1357 (-0.1974, -0.07393) | 1.654956e-05 | log-odds contrast |
| race ethnicity2 | 0.0357209 | 0.05098053 | 0.03572 (-0.0642, 0.1356) | 0.4835045 | log-odds contrast |
| race ethnicity3 | 0.3354555 | 0.1449119 | 0.3355 (0.05143, 0.6195) | 0.02061882 | log-odds contrast |
| race ethnicity4 | -0.08033069 | 0.1428102 | -0.08033 (-0.3602, 0.1996) | 0.5737757 | log-odds contrast |
| race ethnicity5 | 0.227353 | 0.2463188 | 0.2274 (-0.2554, 0.7101) | 0.3560055 | log-odds contrast |
| race ethnicity6 | 0.1810108 | 0.0402531 | 0.181 (0.1021, 0.2599) | 6.897807e-06 | log-odds contrast |
| location1 | -0.04176757 | 0.03056206 | -0.04177 (-0.1017, 0.01813) | 0.1717358 | log-odds contrast |
| location2 | -0.1945384 | 0.03710537 | -0.1945 (-0.2673, -0.1218) | 1.581044e-07 | log-odds contrast |
| location3 | 0.06972893 | 0.0311753 | 0.06973 (0.008626, 0.1308) | 0.02530779 | log-odds contrast |
| sex=0 | unavailable | unavailable | Reference; not estimated | unavailable | reference level |
| race ethnicity=0 | unavailable | unavailable | Reference; not estimated | unavailable | reference level |
| location=0 | unavailable | unavailable | Reference; not estimated | unavailable | reference level |

model id: icd/encounter/spline; note: apparent in-sample diagnostic; no validation or causal claim; analysis population id: primary; population id: primary; outcome id: icd; ascertainment window: encounter; model form: spline.

Table 22: Apparent fit diagnostics

| metric | value |
|---|---|
| log likelihood | -32763.07 |
| null deviance | 68989 |
| residual deviance | 65526.14 |
| AIC | 65556.14 |
| Brier | 0.0453141 |
| log loss | 0.1851837 |
| iterations | 6 |
| residual df | 176907 |
| scaled condition number | 8.864508 |
| events per parameter | 575.6 |
| nonevents per parameter | 11219.2 |

Maximum decrement: 1.

Table 23: Secondary observed-population BMI scenario — panel 1/2

| Model | Reference | Model N | Reference N | Events | Patients shifted | Mean decrement | Eligible (%) |
|---|---|---|---|---|---|---|---|
| abg/encounter/spline | C abg encounter | 29639 | 29639 | 15541 | 29206 | 0.9853909 | 98.54 |
| any gas/encounter/spline | C any gas encounter | 52978 | 52978 | 23621 | 52232 | 0.9859187 | 98.59 |
| icd/encounter/spline | C icd | 176922 | 176922 | 8634 | 174658 | 0.9872034 | 98.72 |

Table 24: Secondary observed-population BMI scenario — panel 2/2

| Model | Observed probability % (95% CI) | Shifted probability % (95% CI) | Difference per 1000 (95% CI) | PIF % (95% CI) |
|---|---|---|---|---|
| abg/encounter/spline | 52.43 (51.87, 52.99) | 51.39 (50.79, 51.99) | 10.42 (8.379, 12.47) | 1.988 (1.597, 2.378) |
| any gas/encounter/ spline | 44.59 (44.17, 45) | 43.74 (43.3, 44.18) | 8.512 (7.054, 9.97) | 1.909 (1.581, 2.236) |
| icd/encounter/spline | 4.88 (4.782, 4.98) | 4.619 (4.521, 4.718) | 2.615 (2.337, 2.893) | 5.359 (4.796, 5.918) |

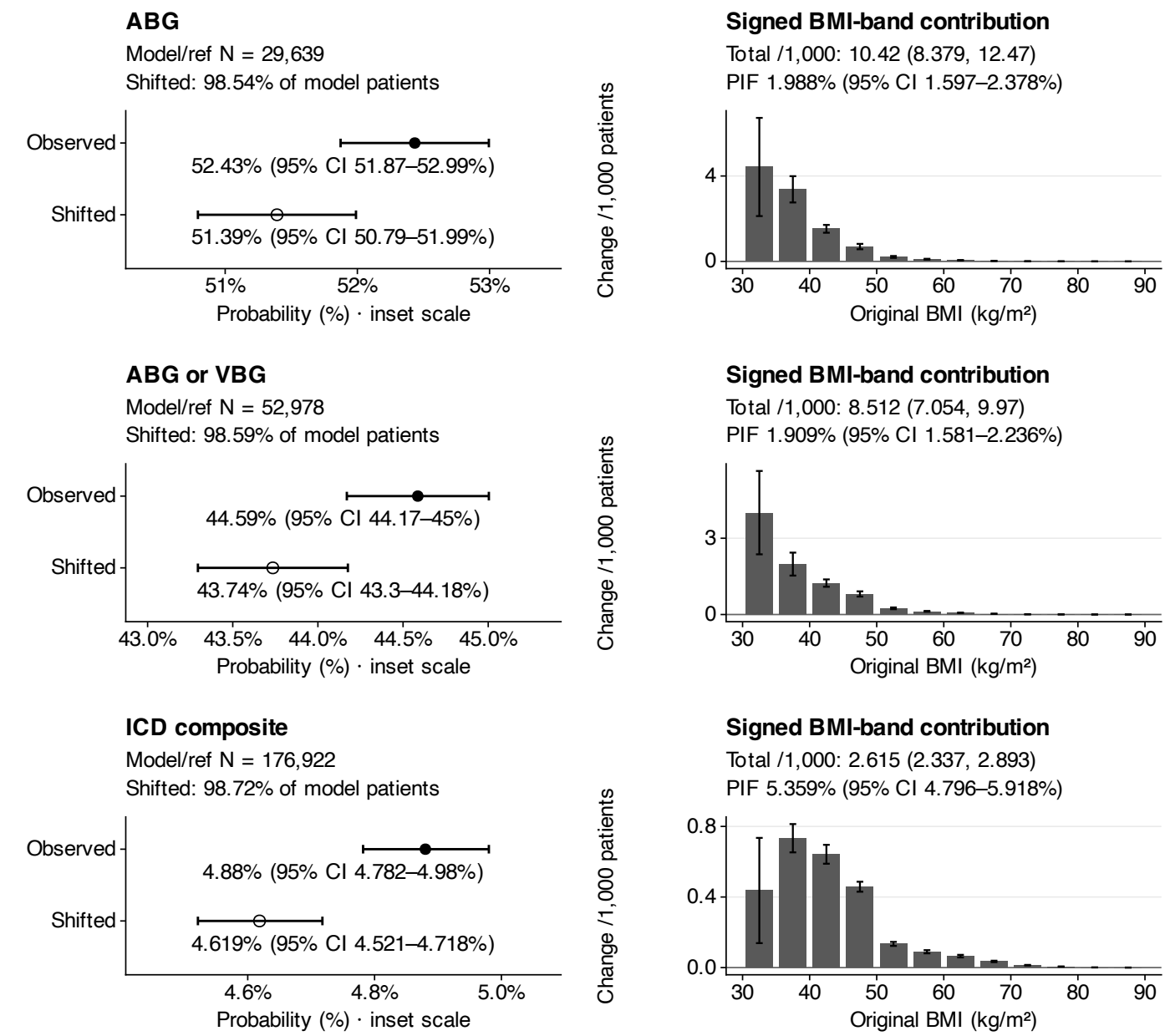



Figure 5: Source-entry population: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F2. Paired scenario: One-unit BMI reduction for observed BMI >30; no floor. ABG: arterial $PCO_2$ ≥45 mm Hg; ABG or VBG: arterial ≥45 or venous ≥50 mm Hg; ICD: five-code composite including E66.2. Left: observed and shifted standardized probabilities with model-based 95% CIs and printed limits. Left probability axes use different labeled inset scales; compare printed values across outcomes. N identifies the common model/reference population; shifted percentage is a descriptive cohort fraction. Right: signed original-BMI-band contributions per 1,000 reference patients, with 95% CIs. Total-difference/PIF limits are printed; contribution covariance reproduces total variance. Right axes vary by outcome and retain zero and all limits. Gas samples are outcome-evaluable; ICD uses known capture. Adjustment, source-entry membership and covariates are fixed. PIF is the model-standardized potential impact fraction. These conditional scenarios do not estimate a weight-loss treatment effect.

fit bmi min: 30; fit bmi max: 89.8000030517578; n observed outside: 0; maximum observed gap: 0; n unseen categories: 0; analysis population id: primary; population id: primary.

Table 25: Curve-grid and transformed support diagnostics — panel 1/2

| model id | n reference | n transformed outside | pct transformed outside | maximum transformed gap | reference id | policy id | method |
|---|---|---|---|---|---|---|---|
| abg/encounter/spline | 29639 | 3181 | 10.73248 | 0.8999996 | C abg encounter | bmi gt30 minus1 v1 | observed |
| any gas/encounter/spline | 52978 | 5548 | 10.47227 | 0.8999996 | C any gas encounter | bmi gt30 minus1 v1 | observed |
| icd/encounter/spline | 176922 | 15514 | 8.768836 | 0.8999996 | C icd | bmi gt30 minus1 v1 | observed |
| abg/index calendar day/spline | 17316 | 1882 | 10.86856 | 0.8999996 | C abg index calendar day | bmi gt30 minus1 v1 | observed |
| any gas/index calendar day/spline | 38877 | 4041 | 10.39432 | 0.8999996 | C any gas index calendar day | bmi gt30 minus1 v1 | observed |
| matched/abg/icd | 29639 | 3181 | 10.73248 | 0.8999996 | M abg | bmi gt30 minus1 v1 | observed |
| matched/abg/abg | 29639 | 3181 | 10.73248 | 0.8999996 | M abg | bmi gt30 minus1 v1 | observed |
| matched/any gas/icd | 52978 | 5548 | 10.47227 | 0.8999996 | M any gas | bmi gt30 minus1 v1 | observed |
| matched/any gas/any gas | 52978 | 5548 | 10.47227 | 0.8999996 | M any gas | bmi gt30 minus1 v1 | observed |
| abg/encounter/spline | 29639 | 0 | 0 | 0 | C abg encounter | none; relationship | fixed BMI curve |
| abg/encounter/linear | 29639 | 0 | 0 | 0 | C abg encounter | none; relationship | fixed BMI curve |
| any gas/encounter/spline | 52978 | 0 | 0 | 0 | C any gas encounter | none; relationship | fixed BMI curve |
| any gas/encounter/linear | 52978 | 0 | 0 | 0 | C any gas encounter | none; relationship | fixed BMI curve |
| icd/encounter/spline | 176922 | 0 | 0 | 0 | C icd | none; relationship | fixed BMI curve |
| icd/encounter/linear | 176922 | 0 | 0 | 0 | C icd | none; relationship | fixed BMI curve |
| abg/index calendar day/spline | 17316 | 0 | 0 | 0 | C abg index calendar day | none; relationship | fixed BMI curve |
| abg/index calendar day/linear | 17316 | 0 | 0 | 0 | C abg index calendar day | none; relationship | fixed BMI curve |
| any gas/index calendar day/spline | 38877 | 0 | 0 | 0 | C any gas index calendar day | none; relationship | fixed BMI curve |
| any gas/index calendar day/linear | 38877 | 0 | 0 | 0 | C any gas index calendar day | none; relationship | fixed BMI curve |
| observation/abg/encounter | 176922 | 0 | 0 | 0 | C | none; relationship | fixed BMI curve |
| observation/vbg/encounter | 176922 | 0 | 0 | 0 | C | none; relationship | fixed BMI curve |
| observation/any gas/encounter | 176922 | 0 | 0 | 0 | C | none; relationship | fixed BMI curve |
| observation/abg/index calendar day | 176922 | 0 | 0 | 0 | C | none; relationship | fixed BMI curve |
| observation/vbg/index calendar day | 176922 | 0 | 0 | 0 | C | none; relationship | fixed BMI curve |
| observation/any gas/index calendar day | 176922 | 0 | 0 | 0 | C | none; relationship | fixed BMI curve |
| matched/abg/icd | 29639 | 0 | 0 | 0 | M abg | none; relationship | fixed BMI curve |
| matched/abg/abg | 29639 | 0 | 0 | 0 | M abg | none; relationship | fixed BMI curve |

| model id | n reference | n transformed outside | pct transformed outside | maximum transformed gap | reference id | policy id | method |
|---|---|---|---|---|---|---|---|
| matched/any gas/icd | 52978 | 0 | 0 | 0 | M any gas | none; relationship | fixed BMI curve |
| matched/any gas/any gas | 52978 | 0 | 0 | 0 | M any gas | none; relationship | fixed BMI curve |

Table 26: Curve-grid and transformed support diagnostics — panel 2/2

| model id | reference population id | outcome id | ascertainment window | model form |
|---|---|---|---|---|
| abg/encounter/spline | C abg encounter | abg | encounter | spline |
| any gas/encounter/spline | C any gas encounter | any gas | encounter | spline |
| icd/encounter/spline | C icd | icd | encounter | spline |
| abg/index calendar day/spline | C abg index calendar day | abg | index calendar day | spline |
| any gas/index calendar day/spline | C any gas index calendar day | any gas | index calendar day | spline |
| matched/abg/icd | M abg | icd | encounter | spline |
| matched/abg/abg | M abg | abg | encounter | spline |
| matched/any gas/icd | M any gas | icd | encounter | spline |
| matched/any gas/any gas | M any gas | any gas | encounter | spline |
| abg/encounter/spline | C abg encounter | abg | encounter | spline |
| abg/encounter/linear | C abg encounter | abg | encounter | linear |
| any gas/encounter/spline | C any gas encounter | any gas | encounter | spline |
| any gas/encounter/linear | C any gas encounter | any gas | encounter | linear |
| icd/encounter/spline | C icd | icd | encounter | spline |
| icd/encounter/linear | C icd | icd | encounter | linear |
| abg/index calendar day/spline | C abg index calendar day | abg | index calendar day | spline |
| abg/index calendar day/linear | C abg index calendar day | abg | index calendar day | linear |
| any gas/index calendar day/spline | C any gas index calendar day | any gas | index calendar day | spline |
| any gas/index calendar day/linear | C any gas index calendar day | any gas | index calendar day | linear |
| observation/abg/encounter | C | abg | encounter | spline |
| observation/vbg/encounter | C | vbg | encounter | spline |
| observation/any gas/encounter | C | any gas | encounter | spline |
| observation/abg/index calendar day | C | abg | index calendar day | spline |
| observation/vbg/index calendar day | C | vbg | index calendar day | spline |
| observation/any gas/index calendar day | C | any gas | index calendar day | spline |
| matched/abg/icd | M abg | icd | encounter | spline |
| matched/abg/abg | M abg | abg | encounter | spline |
| matched/any gas/icd | M any gas | icd | encounter | spline |
| matched/any gas/any gas | M any gas | any gas | encounter | spline |

Table 27: Fixed-index timing transitions

| outcome id | primary | day | n | analysis population id |
|---|---|---|---|---|
| abg | available negative | available negative | 8228 | primary |
| abg | positive | available negative | 1088 | primary |
| abg | unknown | available negative | 0 | primary |
| abg | available negative | positive | 0 | primary |
| abg | positive | positive | 8000 | primary |
| abg | unknown | positive | 0 | primary |

| outcome id | primary | day | n | analysis population id |
|---|---|---|---|---|
| abg | available negative | unknown | 5870 | primary |
| abg | positive | unknown | 6453 | primary |
| abg | unknown | unknown | 147283 | primary |
| any gas | available negative | available negative | 21991 | primary |
| any gas | positive | available negative | 1662 | primary |
| any gas | unknown | available negative | 0 | primary |
| any gas | available negative | positive | 0 | primary |
| any gas | positive | positive | 15224 | primary |
| any gas | unknown | positive | 0 | primary |
| any gas | available negative | unknown | 7366 | primary |
| any gas | positive | unknown | 6735 | primary |
| any gas | unknown | unknown | 123944 | primary |

status: assessed; reason: same-row nested comparison; no model selection; df: 2; analysis population id: primary.

Table 28: Same-sample linear checks

| outcome id | window | n model | likelihood ratio | p value | ascertainment window |
|---|---|---|---|---|---|
| abg | encounter | 29639 | 1.943488 | 0.3784225 | encounter |
| any gas | encounter | 52978 | 6.050963 | 0.04853443 | encounter |
| icd | encounter | 176922 | 15.70311 | 3.891455e-04 | encounter |
| abg | index calendar day | 17316 | 0.1101993 | 0.9463908 | index calendar day |
| any gas | index calendar day | 38877 | 8.900784 | 0.01167399 | index calendar day |

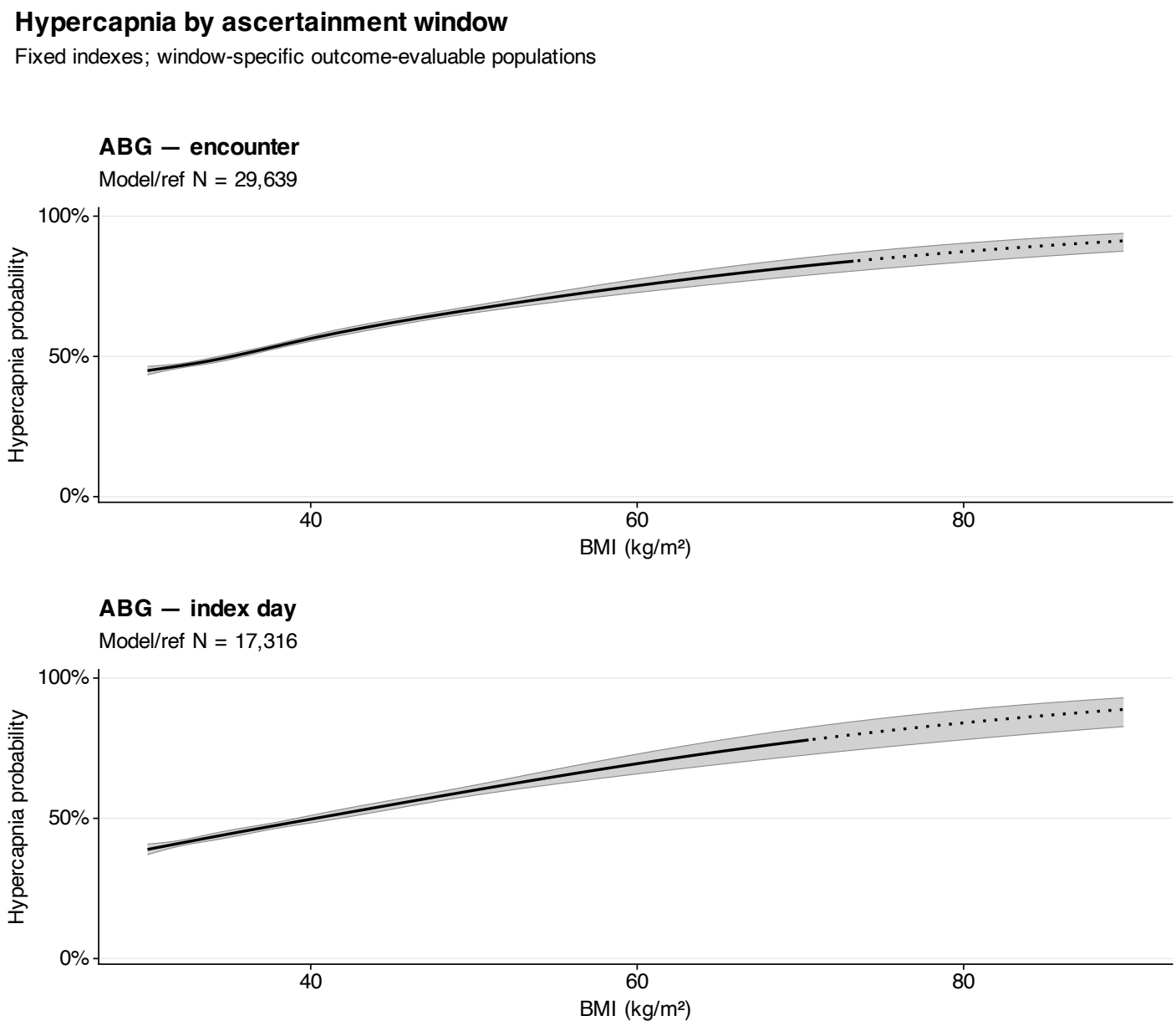



Figure 6: Source-entry population: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F5. Encounter versus encounter-start-calendar-day curves reuse the same indexes and fixed spline basis, on shared BMI/probability scales. Each curve standardizes over its displayed model/reference population. Bands are pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. Pending source or failed models remain explicit. Index day is a calendar-date restriction, not elapsed 24 hours.

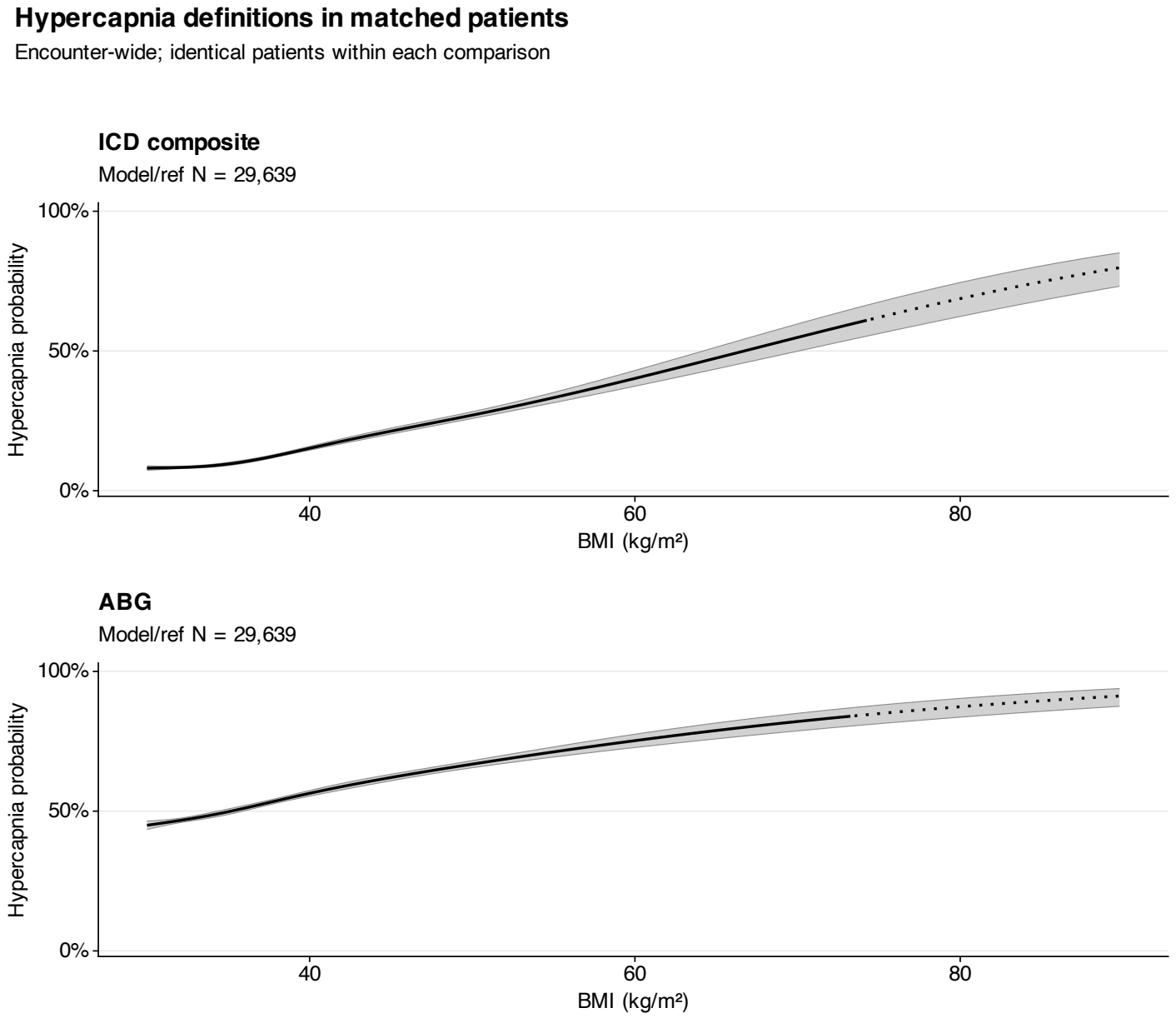



Figure 7: Source-entry population: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F5. Matched ICD/gas curves use identical training/reference patients within each pair and shared BMI/probability scales. Arterial and combined pairs can have different membership. Bands are pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. ICD is the five-code composite including E66.2. Separate bands are not a test of between-model differences.

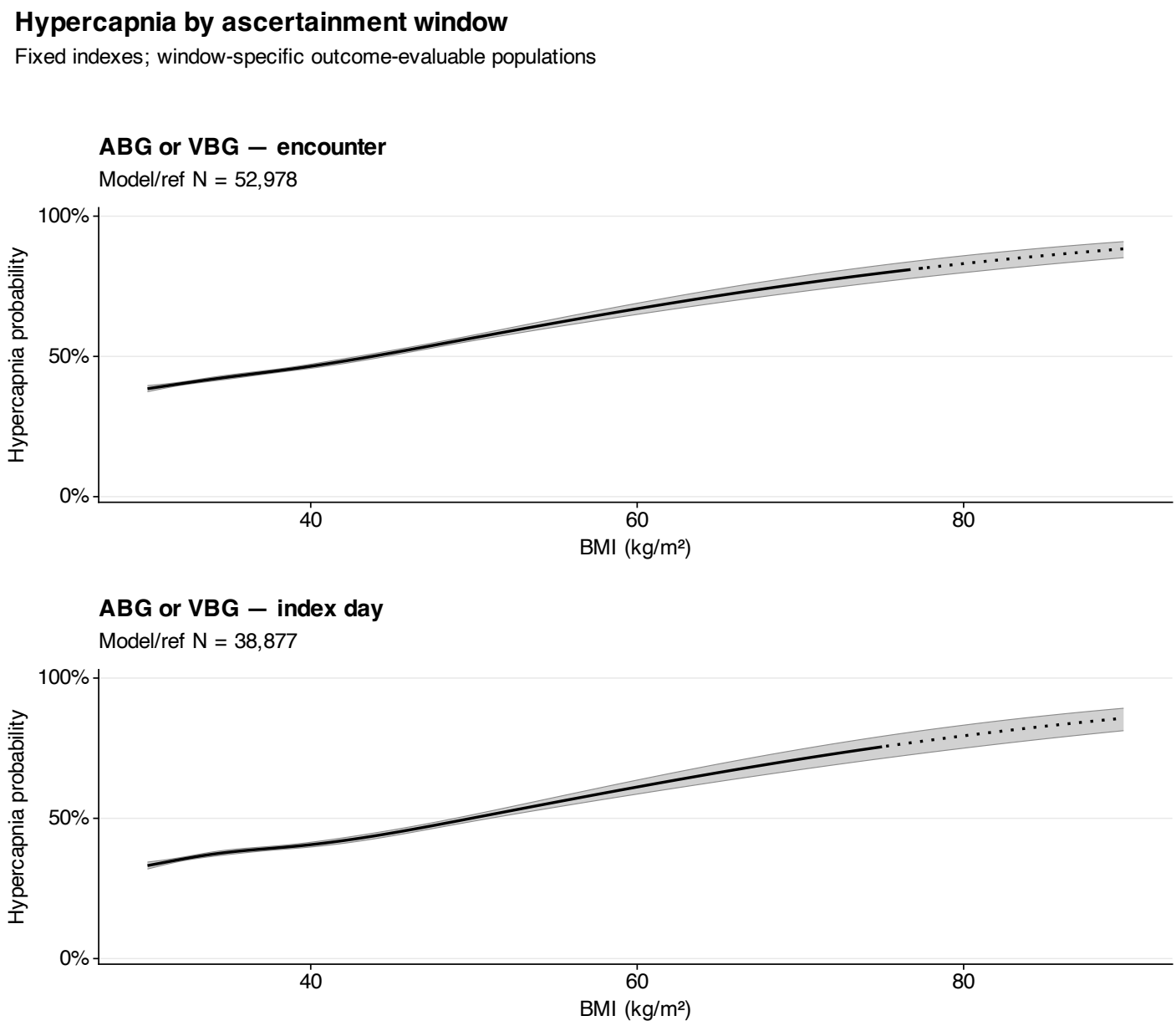



Figure 8: Source-entry population: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F5. Encounter versus encounter-start-calendar-day curves reuse the same indexes and fixed spline basis, on shared BMI/probability scales. Each curve standardizes over its displayed model/reference population. Bands are pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. Pending source or failed models remain explicit. Index day is a calendar-date restriction, not elapsed 24 hours.

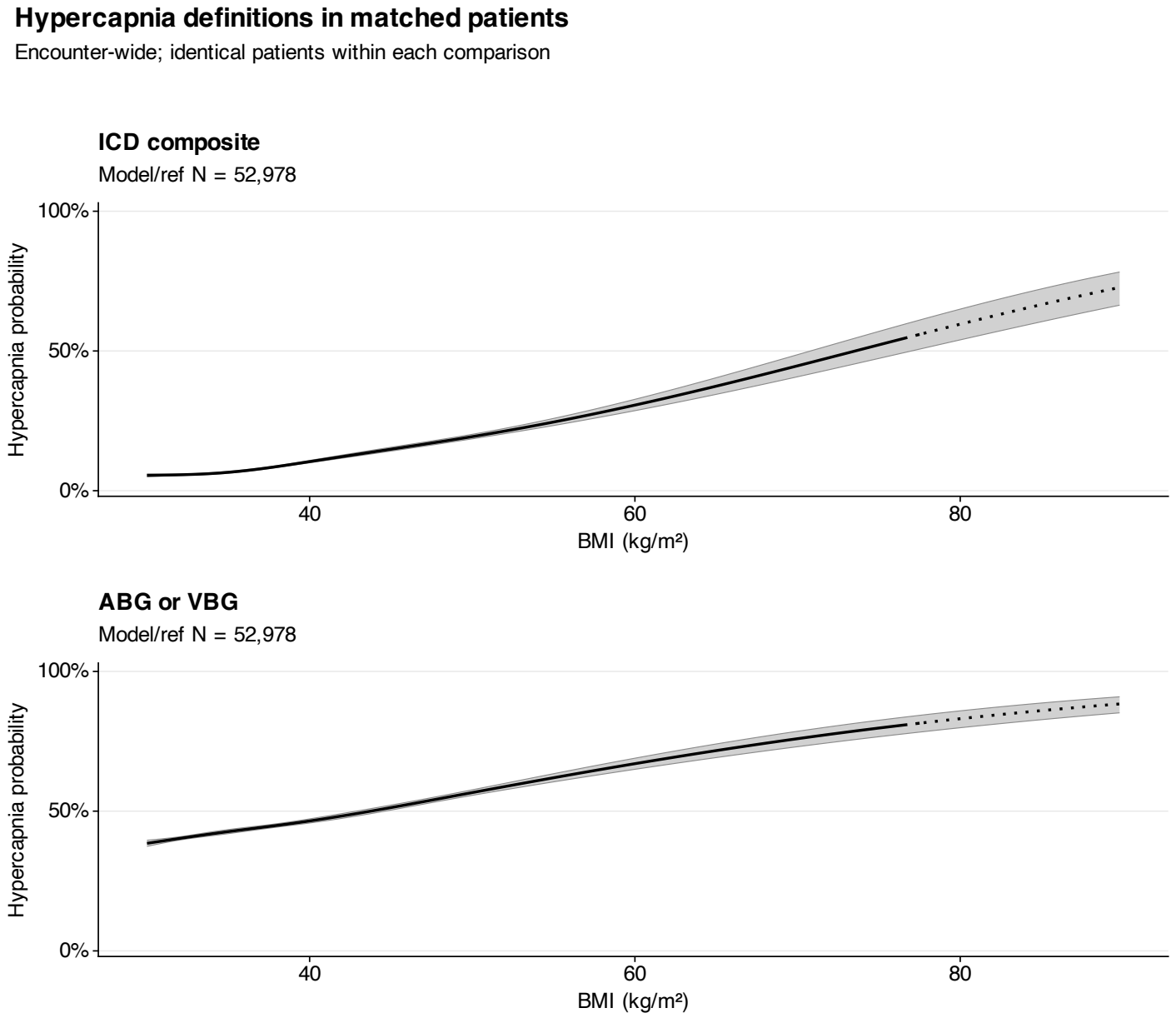



Figure 9: Source-entry population: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F5. Matched ICD/gas curves use identical training/reference patients within each pair and shared BMI/probability scales. Arterial and combined pairs can have different membership. Bands are pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. ICD is the five-code composite including E66.2. Separate bands are not a test of between-model differences.

### 3.3 Measurement provenance and limited current-measurement checks

stage: Q.

Table 29: Primary BMI sources and lags at fixed indexes — panel 1/2

| subgroup | method | bmi band | gas state | metric | n | known | unknown |
|---|---|---|---|---|---|---|---|
| non obesity only | calculated | 40to50 | negative | bmi lag days | 2134 | 0 | 2134 |
| non obesity only | calculated | 40to50 | negative | height lag days | 2134 | 2134 | 0 |
| non obesity only | calculated | 40to50 | negative | weight lag days | 2134 | 2134 | 0 |
| non obesity only | calculated | 40to50 | negative | component gap days | 2134 | 2134 | 0 |
| non obesity only | calculated | 40to50 | positive | bmi lag days | 1967 | 0 | 1967 |
| non obesity only | calculated | 40to50 | positive | height lag days | 1967 | 1967 | 0 |
| non obesity only | calculated | 40to50 | positive | weight lag days | 1967 | 1967 | 0 |
| non obesity only | calculated | 40to50 | positive | component gap days | 1967 | 1967 | 0 |
| non obesity only | calculated | 40to50 | unavailable | bmi lag days | 6299 | 0 | 6299 |
| non obesity only | calculated | 40to50 | unavailable | height lag days | 6299 | 6299 | 0 |
| non obesity only | calculated | 40to50 | unavailable | weight lag days | 6299 | 6299 | 0 |
| non obesity only | calculated | 40to50 | unavailable | component gap days | 6299 | 6299 | 0 |
| non obesity only | calculated | 50plus | negative | bmi lag days | 697 | 0 | 697 |
| non obesity only | calculated | 50plus | negative | height lag days | 697 | 697 | 0 |
| non obesity only | calculated | 50plus | negative | weight lag days | 697 | 697 | 0 |
| non obesity only | calculated | 50plus | negative | component gap days | 697 | 697 | 0 |
| non obesity only | calculated | 50plus | positive | bmi lag days | 1101 | 0 | 1101 |
| non obesity only | calculated | 50plus | positive | height lag days | 1101 | 1101 | 0 |
| non obesity only | calculated | 50plus | positive | weight lag days | 1101 | 1101 | 0 |
| non obesity only | calculated | 50plus | positive | component gap days | 1101 | 1101 | 0 |
| non obesity only | calculated | 50plus | unavailable | bmi lag days | 3576 | 0 | 3576 |
| non obesity only | calculated | 50plus | unavailable | height lag days | 3576 | 3576 | 0 |
| non obesity only | calculated | 50plus | unavailable | weight lag days | 3576 | 3576 | 0 |
| non obesity only | calculated | 50plus | unavailable | component gap days | 3576 | 3576 | 0 |
| non obesity only | calculated | below40 | negative | bmi lag days | 10302 | 0 | 10302 |
| non obesity only | calculated | below40 | negative | height lag days | 10302 | 10302 | 0 |
| non obesity only | calculated | below40 | negative | weight lag days | 10302 | 10302 | 0 |
| non obesity only | calculated | below40 | negative | component gap days | 10302 | 10302 | 0 |
| non obesity only | calculated | below40 | positive | bmi lag days | 7814 | 0 | 7814 |
| non obesity only | calculated | below40 | positive | height lag days | 7814 | 7814 | 0 |
| non obesity only | calculated | below40 | positive | weight lag days | 7814 | 7814 | 0 |
| non obesity only | calculated | below40 | positive | component gap days | 7814 | 7814 | 0 |

| subgroup | method | bmi band | gas state | metric | n | known | unknown |
|---|---|---|---|---|---|---|---|
| non obesity only | calculated | below40 | unavailable | bmi lag days | 21970 | 0 | 21970 |
| non obesity only | calculated | below40 | unavailable | height lag days | 21970 | 21970 | 0 |
| non obesity only | calculated | below40 | unavailable | weight lag days | 21970 | 21970 | 0 |
| non obesity only | calculated | below40 | unavailable | component gap days | 21970 | 21970 | 0 |
| non obesity only | current | 40to50 | negative | bmi lag days | 780 | 780 | 0 |
| non obesity only | current | 40to50 | negative | height lag days | 780 | 761 | 19 |
| non obesity only | current | 40to50 | negative | weight lag days | 780 | 769 | 11 |
| non obesity only | current | 40to50 | negative | component gap days | 780 | 752 | 28 |
| non obesity only | current | 40to50 | positive | bmi lag days | 767 | 767 | 0 |
| non obesity only | current | 40to50 | positive | height lag days | 767 | 739 | 28 |
| non obesity only | current | 40to50 | positive | weight lag days | 767 | 759 | 8 |
| non obesity only | current | 40to50 | positive | component gap days | 767 | 732 | 35 |
| non obesity only | current | 40to50 | unavailable | bmi lag days | 6224 | 6224 | 0 |
| non obesity only | current | 40to50 | unavailable | height lag days | 6224 | 6131 | 93 |
| non obesity only | current | 40to50 | unavailable | weight lag days | 6224 | 6141 | 83 |
| non obesity only | current | 40to50 | unavailable | component gap days | 6224 | 6055 | 169 |
| non obesity only | current | 50plus | negative | bmi lag days | 4 | 4 | 0 |
| non obesity only | current | 50plus | negative | height lag days | 4 | 4 | 0 |
| non obesity only | current | 50plus | negative | weight lag days | 4 | 4 | 0 |
| non obesity only | current | 50plus | negative | component gap days | 4 | 4 | 0 |
| non obesity only | current | 50plus | positive | bmi lag days | 3 | 3 | 0 |
| non obesity only | current | 50plus | positive | height lag days | 3 | 3 | 0 |
| non obesity only | current | 50plus | positive | weight lag days | 3 | 3 | 0 |
| non obesity only | current | 50plus | positive | component gap days | 3 | 3 | 0 |
| non obesity only | current | 50plus | unavailable | bmi lag days | 19 | 19 | 0 |
| non obesity only | current | 50plus | unavailable | height lag days | 19 | 18 | 1 |
| non obesity only | current | 50plus | unavailable | weight lag days | 19 | 19 | 0 |
| non obesity only | current | 50plus | unavailable | component gap days | 19 | 18 | 1 |
| non obesity only | current | below40 | negative | bmi lag days | 8104 | 8104 | 0 |
| non obesity only | current | below40 | negative | height lag days | 8104 | 7685 | 419 |
| non obesity only | current | below40 | negative | weight lag days | 8104 | 7861 | 243 |
| non obesity only | current | below40 | negative | component gap days | 8104 | 7459 | 645 |
| non obesity only | current | below40 | positive | bmi lag days | 6211 | 6211 | 0 |
| non obesity only | current | below40 | positive | height lag days | 6211 | 5952 | 259 |
| non obesity only | current | below40 | positive | weight lag days | 6211 | 6073 | 138 |

| subgroup | method | bmi band | gas state | metric | n | known | unknown |
|---|---|---|---|---|---|---|---|
| non obesity only | current | below40 | positive | component gap days | 6211 | 5825 | 386 |
| non obesity only | current | below40 | unavailable | bmi lag days | 38326 | 38326 | 0 |
| non obesity only | current | below40 | unavailable | height lag days | 38326 | 36873 | 1453 |
| non obesity only | current | below40 | unavailable | weight lag days | 38326 | 37547 | 779 |
| non obesity only | current | below40 | unavailable | component gap days | 38326 | 36147 | 2179 |
| non obesity only | prior | 40to50 | negative | bmi lag days | 876 | 876 | 0 |
| non obesity only | prior | 40to50 | negative | height lag days | 876 | 846 | 30 |
| non obesity only | prior | 40to50 | negative | weight lag days | 876 | 864 | 12 |
| non obesity only | prior | 40to50 | negative | component gap days | 876 | 839 | 37 |
| non obesity only | prior | 40to50 | positive | bmi lag days | 728 | 728 | 0 |
| non obesity only | prior | 40to50 | positive | height lag days | 728 | 713 | 15 |
| non obesity only | prior | 40to50 | positive | weight lag days | 728 | 721 | 7 |
| non obesity only | prior | 40to50 | positive | component gap days | 728 | 708 | 20 |
| non obesity only | prior | 40to50 | unavailable | bmi lag days | 8084 | 8084 | 0 |
| non obesity only | prior | 40to50 | unavailable | height lag days | 8084 | 7501 | 583 |
| non obesity only | prior | 40to50 | unavailable | weight lag days | 8084 | 7994 | 90 |
| non obesity only | prior | 40to50 | unavailable | component gap days | 8084 | 7445 | 639 |
| non obesity only | prior | 50plus | negative | bmi lag days | 12 | 12 | 0 |
| non obesity only | prior | 50plus | negative | height lag days | 12 | 12 | 0 |
| non obesity only | prior | 50plus | negative | weight lag days | 12 | 12 | 0 |
| non obesity only | prior | 50plus | negative | component gap days | 12 | 12 | 0 |
| non obesity only | prior | 50plus | positive | bmi lag days | 15 | 15 | 0 |
| non obesity only | prior | 50plus | positive | height lag days | 15 | 15 | 0 |
| non obesity only | prior | 50plus | positive | weight lag days | 15 | 15 | 0 |
| non obesity only | prior | 50plus | positive | component gap days | 15 | 15 | 0 |
| non obesity only | prior | 50plus | unavailable | bmi lag days | 97 | 97 | 0 |
| non obesity only | prior | 50plus | unavailable | height lag days | 97 | 96 | 1 |
| non obesity only | prior | 50plus | unavailable | weight lag days | 97 | 92 | 5 |
| non obesity only | prior | 50plus | unavailable | component gap days | 97 | 92 | 5 |
| non obesity only | prior | below40 | negative | bmi lag days | 3531 | 3531 | 0 |
| non obesity only | prior | below40 | negative | height lag days | 3531 | 3438 | 93 |
| non obesity only | prior | below40 | negative | weight lag days | 3531 | 3474 | 57 |
| non obesity only | prior | below40 | negative | component gap days | 3531 | 3404 | 127 |
| non obesity only | prior | below40 | positive | bmi lag days | 1763 | 1763 | 0 |
| non obesity only | prior | below40 | positive | height lag days | 1763 | 1718 | 45 |

| subgroup | method | bmi band | gas state | metric | n | known | unknown |
|---|---|---|---|---|---|---|---|
| non obesity only | prior | below40 | positive | weight lag days | 1763 | 1746 | 17 |
| non obesity only | prior | below40 | positive | component gap days | 1763 | 1708 | 55 |
| non obesity only | prior | below40 | unavailable | bmi lag days | 16089 | 16089 | 0 |
| non obesity only | prior | below40 | unavailable | height lag days | 16089 | 14399 | 1690 |
| non obesity only | prior | below40 | unavailable | weight lag days | 16089 | 15842 | 247 |
| non obesity only | prior | below40 | unavailable | component gap days | 16089 | 14219 | 1870 |
| overlap | calculated | 40to50 | negative | bmi lag days | 472 | 0 | 472 |
| overlap | calculated | 40to50 | negative | height lag days | 472 | 472 | 0 |
| overlap | calculated | 40to50 | negative | weight lag days | 472 | 472 | 0 |
| overlap | calculated | 40to50 | negative | component gap days | 472 | 472 | 0 |
| overlap | calculated | 40to50 | positive | bmi lag days | 572 | 0 | 572 |
| overlap | calculated | 40to50 | positive | height lag days | 572 | 572 | 0 |
| overlap | calculated | 40to50 | positive | weight lag days | 572 | 572 | 0 |
| overlap | calculated | 40to50 | positive | component gap days | 572 | 572 | 0 |
| overlap | calculated | 40to50 | unavailable | bmi lag days | 1956 | 0 | 1956 |
| overlap | calculated | 40to50 | unavailable | height lag days | 1956 | 1956 | 0 |
| overlap | calculated | 40to50 | unavailable | weight lag days | 1956 | 1956 | 0 |
| overlap | calculated | 40to50 | unavailable | component gap days | 1956 | 1956 | 0 |
| overlap | calculated | 50plus | negative | bmi lag days | 212 | 0 | 212 |
| overlap | calculated | 50plus | negative | height lag days | 212 | 212 | 0 |
| overlap | calculated | 50plus | negative | weight lag days | 212 | 212 | 0 |
| overlap | calculated | 50plus | negative | component gap days | 212 | 212 | 0 |
| overlap | calculated | 50plus | positive | bmi lag days | 384 | 0 | 384 |
| overlap | calculated | 50plus | positive | height lag days | 384 | 384 | 0 |
| overlap | calculated | 50plus | positive | weight lag days | 384 | 384 | 0 |
| overlap | calculated | 50plus | positive | component gap days | 384 | 384 | 0 |
| overlap | calculated | 50plus | unavailable | bmi lag days | 1666 | 0 | 1666 |
| overlap | calculated | 50plus | unavailable | height lag days | 1666 | 1666 | 0 |
| overlap | calculated | 50plus | unavailable | weight lag days | 1666 | 1666 | 0 |
| overlap | calculated | 50plus | unavailable | component gap days | 1666 | 1666 | 0 |
| overlap | calculated | below40 | negative | bmi lag days | 341 | 0 | 341 |
| overlap | calculated | below40 | negative | height lag days | 341 | 341 | 0 |
| overlap | calculated | below40 | negative | weight lag days | 341 | 341 | 0 |
| overlap | calculated | below40 | negative | component gap days | 341 | 341 | 0 |
| overlap | calculated | below40 | positive | bmi lag days | 390 | 0 | 390 |

| subgroup | method | bmi band | gas state | metric | n | known | unknown |
|---|---|---|---|---|---|---|---|
| overlap | calculated | below40 | positive | height lag days | 390 | 390 | 0 |
| overlap | calculated | below40 | positive | weight lag days | 390 | 390 | 0 |
| overlap | calculated | below40 | positive | component gap days | 390 | 390 | 0 |
| overlap | calculated | below40 | unavailable | bmi lag days | 1167 | 0 | 1167 |
| overlap | calculated | below40 | unavailable | height lag days | 1167 | 1167 | 0 |
| overlap | calculated | below40 | unavailable | weight lag days | 1167 | 1167 | 0 |
| overlap | calculated | below40 | unavailable | component gap days | 1167 | 1167 | 0 |
| overlap | current | 40to50 | negative | bmi lag days | 1284 | 1284 | 0 |
| overlap | current | 40to50 | negative | height lag days | 1284 | 1201 | 83 |
| overlap | current | 40to50 | negative | weight lag days | 1284 | 1235 | 49 |
| overlap | current | 40to50 | negative | component gap days | 1284 | 1154 | 130 |
| overlap | current | 40to50 | positive | bmi lag days | 1203 | 1203 | 0 |
| overlap | current | 40to50 | positive | height lag days | 1203 | 1148 | 55 |
| overlap | current | 40to50 | positive | weight lag days | 1203 | 1176 | 27 |
| overlap | current | 40to50 | positive | component gap days | 1203 | 1124 | 79 |
| overlap | current | 40to50 | unavailable | bmi lag days | 13550 | 13550 | 0 |
| overlap | current | 40to50 | unavailable | height lag days | 13550 | 12771 | 779 |
| overlap | current | 40to50 | unavailable | weight lag days | 13550 | 13235 | 315 |
| overlap | current | 40to50 | unavailable | component gap days | 13550 | 12488 | 1062 |
| overlap | current | 50plus | negative | bmi lag days | 6 | 6 | 0 |
| overlap | current | 50plus | negative | height lag days | 6 | 6 | 0 |
| overlap | current | 50plus | negative | weight lag days | 6 | 5 | 1 |
| overlap | current | 50plus | negative | component gap days | 6 | 5 | 1 |
| overlap | current | 50plus | positive | bmi lag days | 2 | 2 | 0 |
| overlap | current | 50plus | positive | height lag days | 2 | 2 | 0 |
| overlap | current | 50plus | positive | weight lag days | 2 | 2 | 0 |
| overlap | current | 50plus | positive | component gap days | 2 | 2 | 0 |
| overlap | current | 50plus | unavailable | bmi lag days | 49 | 49 | 0 |
| overlap | current | 50plus | unavailable | height lag days | 49 | 47 | 2 |
| overlap | current | 50plus | unavailable | weight lag days | 49 | 47 | 2 |
| overlap | current | 50plus | unavailable | component gap days | 49 | 45 | 4 |
| overlap | current | below40 | negative | bmi lag days | 330 | 330 | 0 |
| overlap | current | below40 | negative | height lag days | 330 | 323 | 7 |
| overlap | current | below40 | negative | weight lag days | 330 | 306 | 24 |
| overlap | current | below40 | negative | component gap days | 330 | 300 | 30 |

| subgroup | method | bmi band | gas state | metric | n | known | unknown |
|---|---|---|---|---|---|---|---|
| overlap | current | below40 | positive | bmi lag days | 434 | 434 | 0 |
| overlap | current | below40 | positive | height lag days | 434 | 426 | 8 |
| overlap | current | below40 | positive | weight lag days | 434 | 402 | 32 |
| overlap | current | below40 | positive | component gap days | 434 | 396 | 38 |
| overlap | current | below40 | unavailable | bmi lag days | 1960 | 1960 | 0 |
| overlap | current | below40 | unavailable | height lag days | 1960 | 1935 | 25 |
| overlap | current | below40 | unavailable | weight lag days | 1960 | 1914 | 46 |
| overlap | current | below40 | unavailable | component gap days | 1960 | 1891 | 69 |
| overlap | prior | 40to50 | negative | bmi lag days | 152 | 152 | 0 |
| overlap | prior | 40to50 | negative | height lag days | 152 | 149 | 3 |
| overlap | prior | 40to50 | negative | weight lag days | 152 | 152 | 0 |
| overlap | prior | 40to50 | negative | component gap days | 152 | 149 | 3 |
| overlap | prior | 40to50 | positive | bmi lag days | 166 | 166 | 0 |
| overlap | prior | 40to50 | positive | height lag days | 166 | 166 | 0 |
| overlap | prior | 40to50 | positive | weight lag days | 166 | 164 | 2 |
| overlap | prior | 40to50 | positive | component gap days | 166 | 164 | 2 |
| overlap | prior | 40to50 | unavailable | bmi lag days | 1723 | 1723 | 0 |
| overlap | prior | 40to50 | unavailable | height lag days | 1723 | 1624 | 99 |
| overlap | prior | 40to50 | unavailable | weight lag days | 1723 | 1718 | 5 |
| overlap | prior | 40to50 | unavailable | component gap days | 1723 | 1623 | 100 |
| overlap | prior | 50plus | negative | bmi lag days | 3 | 3 | 0 |
| overlap | prior | 50plus | negative | height lag days | 3 | 3 | 0 |
| overlap | prior | 50plus | negative | weight lag days | 3 | 3 | 0 |
| overlap | prior | 50plus | negative | component gap days | 3 | 3 | 0 |
| overlap | prior | 50plus | positive | bmi lag days | 1 | 1 | 0 |
| overlap | prior | 50plus | positive | height lag days | 1 | 1 | 0 |
| overlap | prior | 50plus | positive | weight lag days | 1 | 1 | 0 |
| overlap | prior | 50plus | positive | component gap days | 1 | 1 | 0 |
| overlap | prior | 50plus | unavailable | bmi lag days | 28 | 28 | 0 |
| overlap | prior | 50plus | unavailable | height lag days | 28 | 28 | 0 |
| overlap | prior | 50plus | unavailable | weight lag days | 28 | 28 | 0 |
| overlap | prior | 50plus | unavailable | component gap days | 28 | 28 | 0 |
| overlap | prior | below40 | negative | bmi lag days | 117 | 117 | 0 |
| overlap | prior | below40 | negative | height lag days | 117 | 117 | 0 |
| overlap | prior | below40 | negative | weight lag days | 117 | 113 | 4 |

| subgroup | method | bmi band | gas state | metric | n | known | unknown |
|---|---|---|---|---|---|---|---|
| overlap | prior | below40 | negative | component gap days | 117 | 113 | 4 |
| overlap | prior | below40 | positive | bmi lag days | 100 | 100 | 0 |
| overlap | prior | below40 | positive | height lag days | 100 | 99 | 1 |
| overlap | prior | below40 | positive | weight lag days | 100 | 99 | 1 |
| overlap | prior | below40 | positive | component gap days | 100 | 98 | 2 |
| overlap | prior | below40 | unavailable | bmi lag days | 1161 | 1161 | 0 |
| overlap | prior | below40 | unavailable | height lag days | 1161 | 1026 | 135 |
| overlap | prior | below40 | unavailable | weight lag days | 1161 | 1141 | 20 |
| overlap | prior | below40 | unavailable | component gap days | 1161 | 1014 | 147 |

Table 30: Primary BMI sources and lags at fixed indexes — panel 2/2

| subgroup | median | maximum | status |
|---|---|---|---|
| non obesity only | unavailable | unavailable | unavailable or unknown |
| non obesity only | 0 | 6679 | assessed |
| non obesity only | 0 | 6473 | assessed |
| non obesity only | 0 | 6574 | assessed |
| non obesity only | unavailable | unavailable | unavailable or unknown |
| non obesity only | 0 | 3457 | assessed |
| non obesity only | 0 | 5901 | assessed |
| non obesity only | 0 | 5901 | assessed |
| non obesity only | unavailable | unavailable | unavailable or unknown |
| non obesity only | 0 | 7545 | assessed |
| non obesity only | 0 | 7211 | assessed |
| non obesity only | 0 | 7325 | assessed |
| non obesity only | unavailable | unavailable | unavailable or unknown |
| non obesity only | 0 | 4164 | assessed |
| non obesity only | 0 | 6270 | assessed |
| non obesity only | 0 | 6255 | assessed |
| non obesity only | unavailable | unavailable | unavailable or unknown |
| non obesity only | 0 | 6641 | assessed |
| non obesity only | 0 | 6641 | assessed |
| non obesity only | 0 | 5160 | assessed |
| non obesity only | unavailable | unavailable | unavailable or unknown |
| non obesity only | 0 | 6817 | assessed |
| non obesity only | 0 | 7830 | assessed |
| non obesity only | 0 | 7830 | assessed |
| non obesity only | unavailable | unavailable | unavailable or unknown |
| non obesity only | 0 | 7064 | assessed |
| non obesity only | 0 | 6901 | assessed |
| non obesity only | 0 | 7064 | assessed |
| non obesity only | unavailable | unavailable | unavailable or unknown |
| non obesity only | 0 | 6433 | assessed |
| non obesity only | 0 | 7318 | assessed |
| non obesity only | 0 | 6593 | assessed |
| non obesity only | unavailable | unavailable | unavailable or unknown |
| non obesity only | 0 | 8039 | assessed |
| non obesity only | 0 | 6935 | assessed |
| non obesity only | 0 | 8039 | assessed |
| non obesity only | 0 | 0 | assessed |
| non obesity only | 0 | 2717 | assessed |
| non obesity only | 0 | 4165 | assessed |
| non obesity only | 0 | 4170 | assessed |
| non obesity only | 0 | 0 | assessed |
| non obesity only | 0 | 2646 | assessed |
| non obesity only | 0 | 3567 | assessed |
| non obesity only | 0 | 3569 | assessed |
| non obesity only | 0 | 0 | assessed |
| non obesity only | 0 | 6469 | assessed |

| subgroup | median | maximum | status |
|---|---|---|---|
| non obesity only | 0 | 4192 | assessed |
| non obesity only | 0 | 6469 | assessed |
| non obesity only | -5 | 0 | assessed |
| non obesity only | 0 | 1749 | assessed |
| non obesity only | 0 | 0 | assessed |
| non obesity only | 4.5 | 1749 | assessed |
| non obesity only | 0 | 0 | assessed |
| non obesity only | 0 | 0 | assessed |
| non obesity only | 0 | 0 | assessed |
| non obesity only | 0 | 2 | assessed |
| non obesity only | 0 | 0 | assessed |
| non obesity only | 0 | 2 | assessed |
| non obesity only | 0 | 409 | assessed |
| non obesity only | 0 | 409 | assessed |
| non obesity only | 0 | 0 | assessed |
| non obesity only | 0 | 6685 | assessed |
| non obesity only | 0 | 4250 | assessed |
| non obesity only | 0 | 6685 | assessed |
| non obesity only | 0 | 0 | assessed |
| non obesity only | 0 | 6658 | assessed |
| non obesity only | 0 | 4372 | assessed |
| non obesity only | 0 | 6658 | assessed |
| non obesity only | 0 | 0 | assessed |
| non obesity only | 0 | 6984 | assessed |
| non obesity only | 0 | 4399 | assessed |
| non obesity only | 0 | 6984 | assessed |
| non obesity only | 190.5 | 6240 | assessed |
| non obesity only | 16 | 6240 | assessed |
| non obesity only | 3 | 5376 | assessed |
| non obesity only | 6 | 6119 | assessed |
| non obesity only | 186.5 | 6524 | assessed |
| non obesity only | 4 | 6641 | assessed |
| non obesity only | 1 | 4964 | assessed |
| non obesity only | 0 | 6482 | assessed |
| non obesity only | 141 | 6888 | assessed |
| non obesity only | 29 | 6756 | assessed |
| non obesity only | 16 | 6609 | assessed |
| non obesity only | 0 | 6756 | assessed |
| non obesity only | 201.5 | 2304 | assessed |
| non obesity only | 0 | 467 | assessed |
| non obesity only | 0 | 349 | assessed |
| non obesity only | 0 | 429 | assessed |
| non obesity only | 382 | 4581 | assessed |
| non obesity only | 1 | 1072 | assessed |
| non obesity only | 1 | 1072 | assessed |
| non obesity only | 0 | 326 | assessed |
| non obesity only | 357 | 3937 | assessed |
| non obesity only | 0 | 826 | assessed |
| non obesity only | 0 | 2534 | assessed |
| non obesity only | 0 | 2534 | assessed |
| non obesity only | 132 | 5223 | assessed |
| non obesity only | 21 | 5223 | assessed |
| non obesity only | 6 | 6669 | assessed |
| non obesity only | 15 | 5515 | assessed |
| non obesity only | 93 | 7231 | assessed |
| non obesity only | 11 | 5934 | assessed |
| non obesity only | 10 | 5934 | assessed |
| non obesity only | 1 | 4677 | assessed |
| non obesity only | 105 | 7729 | assessed |
| non obesity only | 62 | 7218 | assessed |
| non obesity only | 38 | 7729 | assessed |
| non obesity only | 0 | 6069 | assessed |
| overlap | unavailable | unavailable | unavailable or unknown |
| overlap | 0 | 4039 | assessed |
| overlap | 0 | 5859 | assessed |
| overlap | 0 | 5859 | assessed |
| overlap | unavailable | unavailable | unavailable or unknown |
| overlap | 0 | 4056 | assessed |
| overlap | 0 | 5712 | assessed |
| overlap | 0 | 5712 | assessed |
| overlap | unavailable | unavailable | unavailable or unknown |

| subgroup | median | maximum | status |
|---|---|---|---|
| overlap | 0 | 5701 | assessed |
| overlap | 0 | 6495 | assessed |
| overlap | 0 | 6441 | assessed |
| overlap | unavailable | unavailable | unavailable or unknown |
| overlap | 0 | 1761 | assessed |
| overlap | 0 | 5560 | assessed |
| overlap | 0 | 5560 | assessed |
| overlap | unavailable | unavailable | unavailable or unknown |
| overlap | 0 | 2286 | assessed |
| overlap | 0 | 5599 | assessed |
| overlap | 0 | 5599 | assessed |
| overlap | unavailable | unavailable | unavailable or unknown |
| overlap | 0 | 4505 | assessed |
| overlap | 0 | 5605 | assessed |
| overlap | 0 | 5605 | assessed |
| overlap | unavailable | unavailable | unavailable or unknown |
| overlap | 0 | 4404 | assessed |
| overlap | 0 | 5871 | assessed |
| overlap | 0 | 5871 | assessed |
| overlap | unavailable | unavailable | unavailable or unknown |
| overlap | 0 | 3386 | assessed |
| overlap | 0 | 5687 | assessed |
| overlap | 2 | 5687 | assessed |
| overlap | unavailable | unavailable | unavailable or unknown |
| overlap | 0 | 4144 | assessed |
| overlap | 0 | 5902 | assessed |
| overlap | 0 | 5903 | assessed |
| overlap | 0 | 0 | assessed |
| overlap | 0 | 6618 | assessed |
| overlap | 0 | 3739 | assessed |
| overlap | 0 | 6618 | assessed |
| overlap | 0 | 0 | assessed |
| overlap | 0 | 4569 | assessed |
| overlap | 0 | 3154 | assessed |
| overlap | 0 | 4569 | assessed |
| overlap | 0 | 0 | assessed |
| overlap | 0 | 6802 | assessed |
| overlap | 0 | 4434 | assessed |
| overlap | 0 | 6802 | assessed |
| overlap | 0 | 0 | assessed |
| overlap | 0 | 0 | assessed |
| overlap | 0 | 0 | assessed |
| overlap | 0 | 0 | assessed |
| overlap | 0 | 0 | assessed |
| overlap | 0 | 0 | assessed |
| overlap | 0 | 0 | assessed |
| overlap | 0 | 0 | assessed |
| overlap | 0 | 0 | assessed |
| overlap | 0 | 2322 | assessed |
| overlap | 0 | 0 | assessed |
| overlap | 0 | 2322 | assessed |
| overlap | 0 | 0 | assessed |
| overlap | 0 | 2798 | assessed |
| overlap | 0 | 3909 | assessed |
| overlap | 0 | 3909 | assessed |
| overlap | 0 | 0 | assessed |
| overlap | 0 | 2750 | assessed |
| overlap | 0 | 3844 | assessed |
| overlap | 0 | 3844 | assessed |
| overlap | 0 | 0 | assessed |
| overlap | 0 | 6976 | assessed |
| overlap | 0 | 3962 | assessed |
| overlap | 0 | 6976 | assessed |
| overlap | 533.5 | 4797 | assessed |
| overlap | 0 | 3649 | assessed |
| overlap | 168.5 | 3477 | assessed |
| overlap | 163 | 3289 | assessed |
| overlap | 583 | 5694 | assessed |
| overlap | 0 | 3227 | assessed |
| overlap | 0 | 3595 | assessed |
| overlap | 2 | 3595 | assessed |

| subgroup | median | maximum | status |
|---|---|---|---|
| overlap | 489 | 7039 | assessed |
| overlap | 59 | 6769 | assessed |
| overlap | 100 | 6455 | assessed |
| overlap | 0 | 4835 | assessed |
| overlap | 149 | 681 | assessed |
| overlap | 0 | 149 | assessed |
| overlap | 75 | 149 | assessed |
| overlap | 1 | 75 | assessed |
| overlap | 190 | 190 | assessed |
| overlap | 0 | 0 | assessed |
| overlap | 0 | 0 | assessed |
| overlap | 0 | 0 | assessed |
| overlap | 1060 | 4159 | assessed |
| overlap | 0 | 2421 | assessed |
| overlap | 1 | 1648 | assessed |
| overlap | 0 | 1659 | assessed |
| overlap | 303 | 4649 | assessed |
| overlap | 0 | 3485 | assessed |
| overlap | 148 | 3360 | assessed |
| overlap | 84 | 3360 | assessed |
| overlap | 325 | 3698 | assessed |
| overlap | 0 | 3698 | assessed |
| overlap | 134 | 3698 | assessed |
| overlap | 39 | 2303 | assessed |
| overlap | 135 | 6981 | assessed |
| overlap | 50.5 | 6981 | assessed |
| overlap | 63 | 6172 | assessed |
| overlap | 0 | 4425 | assessed |

### 3.3.1 primary_current

family: supporting; window: encounter; model form: spline; source status: qualified; formula text: .outcome ~ bmi_ns1 + bmi_ns2 + bmi_ns3 + age_at_encounter + sex + race_ethnicity + location; basis id: 026b7a1829cd49840558b84e972374c0e1cb1f3a2f2304811582736b3d37f105; weight specification: none; covariance method: model-based binomial; empirical reference fixed; n eligible: 126480; n excluded adjustment: 0; n parameters: 15; model rank: 15; status: successful; failure reason: ; warnings: ; analysis population id: primary_current; ascertainment window: encounter; population id: primary_current; source reason: unavailable; estimator: unavailable; separation detected: unavailable.

Table 31: Current-measurement fit status — panel 1/2

| model id | outcome id | training population id | field | n excluded outcome | n model | n patients | n events |
|---|---|---|---|---|---|---|---|
| abg/encounter/spline | abg | C abg encounter | abg | 103429 | 23051 | 23051 | 12201 |
| any gas/encounter/spline | any gas | C any gas encounter | any gas | 87865 | 38615 | 38615 | 17769 |
| icd/encounter/spline | icd | C icd | icd | 0 | 126480 | 126480 | 6258 |

Table 32: Current-measurement fit status — panel 2/2

| model id | n nonevents | fit runtime |
|---|---|---|
| abg/encounter/spline | 10850 | 0.098 |
| any gas/encounter/spline | 20846 | 0.16 |
| icd/encounter/spline | 120222 | 0.563 |

Maximum decrement: 1.

Table 33: Current-measurement scenario — panel 1/2

| Model | Reference | Model N | Reference N | Events | Patients shifted | Mean decrement | Eligible (%) |
|---|---|---|---|---|---|---|---|
| abg/encounter/spline | C abg encounter | 23051 | 23051 | 12201 | 22714 | 0.9853802 | 98.54 |
| any gas/encounter/ spline | C any gas encounter | 38615 | 38615 | 17769 | 38053 | 0.9854461 | 98.54 |
| icd/encounter/spline | C icd | 126480 | 126480 | 6258 | 124789 | 0.9866303 | 98.66 |

Table 34: Current-measurement scenario — panel 2/2

| Model | Observed probability % (95% CI) | Shifted probability % (95% CI) | Difference per 1000 (95% CI) | PIF % (95% CI) |
|---|---|---|---|---|
| abg/encounter/spline | 52.93 (52.29, 53.57) | 51.85 (51.17, 52.53) | 10.77 (8.423, 13.12) | 2.035 (1.589, 2.478) |
| any gas/encounter/ spline | 46.02 (45.53, 46.51) | 45.12 (44.61, 45.64) | 8.917 (7.182, 10.65) | 1.938 (1.56, 2.315) |
| icd/encounter/spline | 4.948 (4.831, 5.067) | 4.688 (4.572, 4.806) | 2.6 (2.263, 2.936) | 5.254 (4.582, 5.922) |

## 3.4 Same-day arterial $PCO_2$/pH co-occurrence

This secondary analysis uses any recorded calendar date within the existing index encounter with both usable ordinary arterial $PCO_2$ and arterial pH. It is **same recorded-day co-occurrence; not verified same-sample pairing**. The export dates when results were recorded; it does not identify a shared blood sample, collection time or elapsed 24-hour interval. Validity screens are $5<PCO_2<200$ mm Hg and $6.5<pH<7.8$, applied before display rounding.

The source endpoints use the same patients with at least one co-observed day: any $PaCO_2 \geq 45$ on such a day, and any such day also containing pH <7.35. Results may come from different samples. Elevated $PCO_2$ on one date and low pH on another cannot establish co-occurrence. Patients without a valid co-observed day have unknown supplemental outcomes. Additional unusable or single-analyte records remain audited without erasing usable days. A negative describes eligible recorded evidence only. The primary numeric BMI spline basis, inherited adjustment and common supplemental reference population are retained. A single unpenalized multinomial logistic spline assigns each patient to no documented hypercapnia, hypercapnia without documented same-day acidemia, or hypercapnia with same-day acidemia. Any eligible hypercapnia–acidemia day takes priority, including patients with other hypercapnic days without acidemia. The baseline outcome is no documented hypercapnia. All three probabilities are standardized over the same patients; full cross-category covariance supplies pointwise intervals and comparisons of component changes. No additional PIF is calculated.

The full observed BMI range is the main relationship, with sparse regions flagged. BMI 40–45 is a prespecified numerical summary, not a biological cutoff or fitting restriction. Every supported five-unit interval is also reported. These are descriptive pointwise contrasts without multiplicity adjustment; separate significance counts do not establish differences in associations.

Table 35: Recorded-day source qualification

| module | status | reason | n primary complete | n coobserved | analysis population id |
|---|---|---|---|---|---|
| same day arterial cooccurrence v1 | qualified | Recorded-day evidence assessed; zeros refer only to co-observed days | 176922 | 22195 | primary |

Table 36: Four-stage patient/evidence reconciliation; raw proportions with Wilson 95% intervals

| Stage | Patients | Events | Observed | Status |
|---|---|---|---|---|
| Original arterial-evaluable | 29639 | 15541 | 52.43% (95% CI 51.87–53.00%) | assessed |
| Co-observed patients: original arterial outcome | 22195 | 11690 | 52.67% (95% CI 52.01–53.33%) | qualified |
| Same patients: hypercapnia on co-observed days | 22195 | 11669 | 52.57% (95% CI 51.92–53.23%) | qualified |
| Same patients: same-day hypercapnia and acidemia | 22195 | 8809 | 39.69% (95% CI 39.05–40.33%) | qualified |

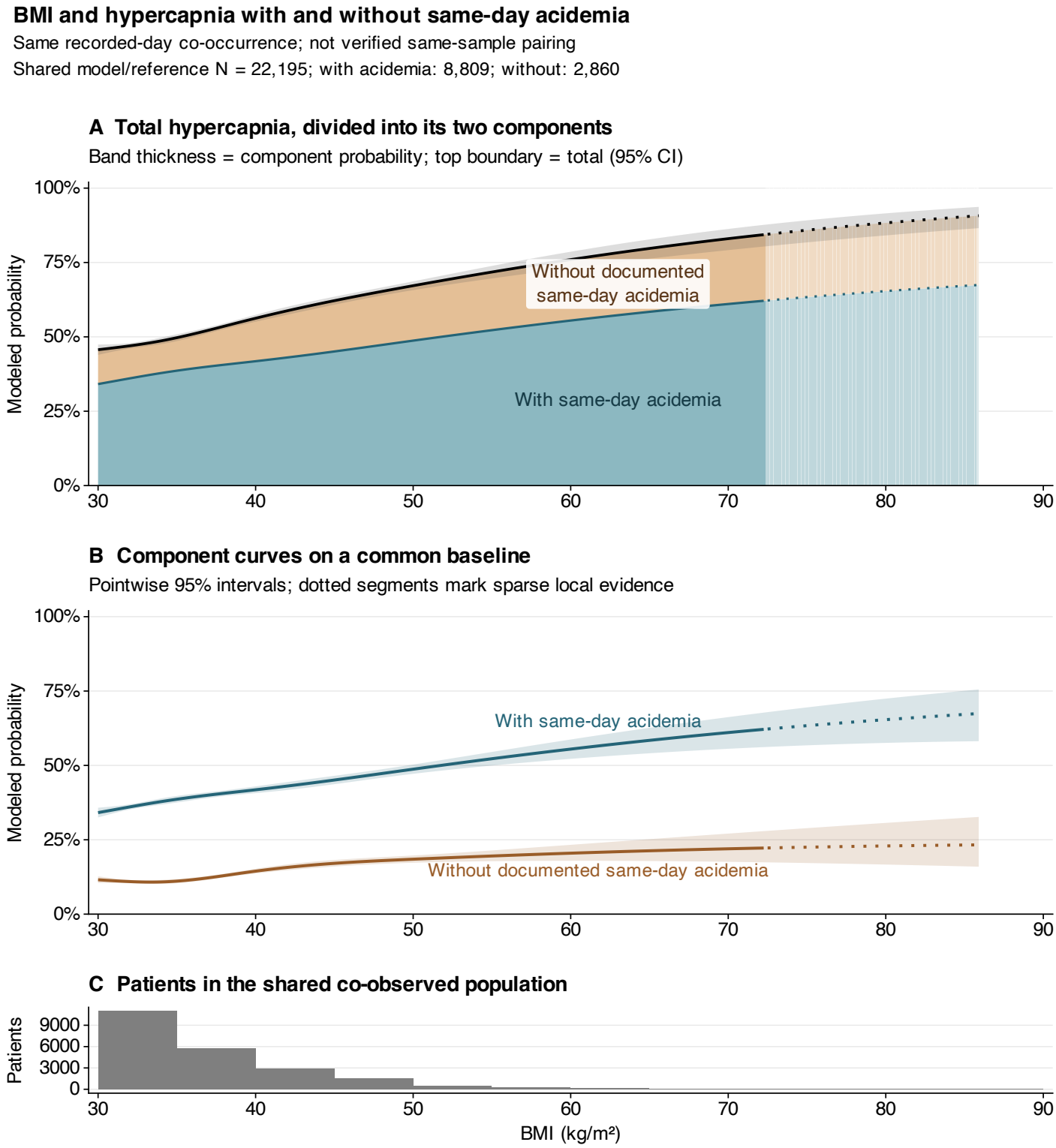



Figure 10: Same recorded-day co-occurrence; not verified same-sample pairing. Primary fixed-index patients with both arterial analytes share inherited adjustment and the primary BMI spline. Any eligible day with $PaCO_2$ ≥45 mm Hg and pH <7.35 assigns the with-acidemia category; remaining hypercapnic patients have no documented same-day acidemia. Different samples may contribute. A: band thicknesses give component probabilities; the upper boundary is total hypercapnia. B: components share a baseline. Shading denotes pointwise 95% intervals using full joint covariance. C: shared-population BMI frequencies. Fading/dotted lines mark local windows below 30 patients or five events/non-events; marginal checks do not establish joint support. These recorded outcomes do not establish simultaneous physiology, acute respiratory failure or compensation. Selection, treatment timing and confounding remain limitations.

Table 37: Mutually exclusive patient categories; any acidemia co-occurrence takes priority

| category id | n | N | status | definition | analysis population id |
|---|---|---|---|---|---|
| no hypercapnia | 10526 | 22195 | qualified | No documented hypercapnia on co-observed days | primary |
| without acidemia | 2860 | 22195 | qualified | Hypercapnia without documented same-day acidemia on any eligible day | primary |
| with acidemia | 8809 | 22195 | qualified | Any eligible day with hypercapnia and acidemia; takes priority over other days | primary |

Table 38: Component magnitudes across BMI: absolute changes in percentage points and probability ratios, with 95% intervals

| BMI interval | Outcome | Lower BMI probability % (95% CI) | Higher BMI probability % (95% CI) | Change pp (95% CI) | Probability ratio (95% CI) | Status |
|---|---|---|---|---|---|---|
| 30 to 35 | No documented hypercapnia | 54.32 (52.58 to 56.06) | 50.28 (49.14 to 51.42) | -4.04 (-6.42 to -1.67) | 0.93 (0.88 to 0.97) | Estimated |
| 30 to 35 | Hypercapnia without documented same-day acidemia | 11.55 (10.48 to 12.71) | 11.10 (10.41 to 11.82) | -0.45 (-1.95 to 1.04) | 0.96 (0.84 to 1.10) | Estimated |
| 30 to 35 | Hypercapnia with same-day acidemia | 34.13 (32.51 to 35.79) | 38.62 (37.53 to 39.73) | 4.50 (2.24 to 6.75) | 1.13 (1.06 to 1.21) | Estimated |
| 30 to 35 | Any hypercapnia | 45.68 (43.94 to 47.42) | 49.72 (48.58 to 50.86) | 4.04 (1.67 to 6.42) | 1.09 (1.03 to 1.14) | Estimated |
| 30 to 35 | With minus without: difference in changes | Not estimated | Not estimated | 4.95 (1.94 to 7.96) | Not estimated | Estimated |
| 35 to 40 | No documented hypercapnia | 50.28 (49.14 to 51.42) | 43.74 (42.58 to 44.91) | -6.54 (-8.05 to -5.02) | 0.87 (0.84 to 0.90) | Estimated |
| 35 to 40 | Hypercapnia without documented same-day acidemia | 11.10 (10.41 to 11.82) | 14.45 (13.68 to 15.26) | 3.35 (2.40 to 4.30) | 1.30 (1.21 to 1.40) | Estimated |
| 35 to 40 | Hypercapnia with same-day acidemia | 38.62 (37.53 to 39.73) | 41.81 (40.67 to 42.95) | 3.18 (1.72 to 4.64) | 1.08 (1.04 to 1.12) | Estimated |
| 35 to 40 | Any hypercapnia | 49.72 (48.58 to 50.86) | 56.26 (55.09 to 57.42) | 6.54 (5.02 to 8.05) | 1.13 (1.10 to 1.16) | Estimated |
| 35 to 40 | With minus without: difference in changes | Not estimated | Not estimated | -0.17 (-2.12 to 1.77) | Not estimated | Estimated |
| 40 to 45 | No documented hypercapnia | 43.74 (42.58 to 44.91) | 37.83 (36.49 to 39.19) | -5.91 (-6.63 to -5.19) | 0.86 (0.85 to 0.88) | Estimated |
| 40 to 45 | Hypercapnia without documented same-day acidemia | 14.45 (13.68 to 15.26) | 17.08 (16.01 to 18.21) | 2.63 (2.03 to 3.23) | 1.18 (1.14 to 1.22) | Estimated |
| 40 to 45 | Hypercapnia with same-day acidemia | 41.81 (40.67 to 42.95) | 45.09 (43.68 to 46.50) | 3.28 (2.54 to 4.02) | 1.08 (1.06 to 1.10) | Estimated |
| 40 to 45 | Any hypercapnia | 56.26 (55.09 to 57.42) | 62.17 (60.81 to 63.51) | 5.91 (5.19 to 6.63) | 1.11 (1.09 to 1.12) | Estimated |
| 40 to 45 | With minus without: difference in changes | Not estimated | Not estimated | 0.65 (-0.49 to 1.79) | Not estimated | Estimated |
| 45 to 50 | No documented hypercapnia | 37.83 (36.49 to 39.19) | 32.80 (31.34 to 34.29) | -5.03 (-6.11 to -3.96) | 0.87 (0.84 to 0.89) | Estimated |
| 45 to 50 | Hypercapnia without documented same-day acidemia | 17.08 (16.01 to 18.21) | 18.47 (17.28 to 19.72) | 1.39 (0.60 to 2.18) | 1.08 (1.03 to 1.13) | Estimated |
| 45 to 50 | Hypercapnia with same-day acidemia | 45.09 (43.68 to 46.50) | 48.73 (47.17 to 50.29) | 3.64 (2.58 to 4.71) | 1.08 (1.06 to 1.11) | Estimated |
| 45 to 50 | Any hypercapnia | 62.17 (60.81 to 63.51) | 67.20 (65.71 to 68.66) | 5.03 (3.96 to 6.11) | 1.08 (1.06 to 1.10) | Estimated |
| 45 to 50 | With minus without: difference in changes | Not estimated | Not estimated | 2.25 (0.72 to 3.79) | Not estimated | Estimated |

| BMI interval | Outcome | Lower BMI probability % (95% CI) | Higher BMI probability % (95% CI) | Change pp (95% CI) | Probability ratio (95% CI) | Status |
|---|---|---|---|---|---|---|
| 50 to 55 | No documented hypercapnia | 32.80 (31.34 to 34.29) | 28.21 (26.15 to 30.36) | -4.59 (-5.61 to -3.58) | 0.86 (0.83 to 0.89) | Estimated |
| 50 to 55 | Hypercapnia without documented same-day acidemia | 18.47 (17.28 to 19.72) | 19.55 (17.80 to 21.42) | 1.08 (0.12 to 2.04) | 1.06 (1.01 to 1.11) | Estimated |
| 50 to 55 | Hypercapnia with same-day acidemia | 48.73 (47.17 to 50.29) | 52.25 (49.95 to 54.53) | 3.51 (2.34 to 4.69) | 1.07 (1.05 to 1.10) | Estimated |
| 50 to 55 | Any hypercapnia | 67.20 (65.71 to 68.66) | 71.79 (69.64 to 73.85) | 4.59 (3.58 to 5.61) | 1.07 (1.05 to 1.08) | Estimated |
| 50 to 55 | With minus without: difference in changes | Not estimated | Not estimated | 2.44 (0.55 to 4.32) | Not estimated | Estimated |
| 55 to 60 | No documented hypercapnia | 28.21 (26.15 to 30.36) | 24.02 (21.38 to 26.88) | -4.18 (-4.99 to -3.38) | 0.85 (0.82 to 0.89) | Estimated |
| 55 to 60 | Hypercapnia without documented same-day acidemia | 19.55 (17.80 to 21.42) | 20.49 (17.94 to 23.30) | 0.95 (-0.08 to 1.97) | 1.05 (1.00 to 1.10) | Estimated |
| 55 to 60 | Hypercapnia with same-day acidemia | 52.25 (49.95 to 54.53) | 55.48 (52.20 to 58.72) | 3.24 (2.07 to 4.40) | 1.06 (1.04 to 1.08) | Estimated |
| 55 to 60 | Any hypercapnia | 71.79 (69.64 to 73.85) | 75.98 (73.12 to 78.62) | 4.18 (3.38 to 4.99) | 1.06 (1.05 to 1.07) | Estimated |
| 55 to 60 | With minus without: difference in changes | Not estimated | Not estimated | 2.29 (0.25 to 4.34) | Not estimated | Estimated |
| 60 to 65 | No documented hypercapnia | 24.02 (21.38 to 26.88) | 20.28 (17.23 to 23.72) | -3.74 (-4.32 to -3.16) | 0.84 (0.81 to 0.89) | Estimated |
| 60 to 65 | Hypercapnia without documented same-day acidemia | 20.49 (17.94 to 23.30) | 21.30 (17.85 to 25.21) | 0.81 (-0.28 to 1.89) | 1.04 (0.99 to 1.09) | Estimated |
| 60 to 65 | Hypercapnia with same-day acidemia | 55.48 (52.20 to 58.72) | 58.42 (54.07 to 62.64) | 2.94 (1.77 to 4.10) | 1.05 (1.03 to 1.07) | Estimated |
| 60 to 65 | Any hypercapnia | 75.98 (73.12 to 78.62) | 79.72 (76.28 to 82.77) | 3.74 (3.16 to 4.32) | 1.05 (1.04 to 1.06) | Estimated |
| 60 to 65 | With minus without: difference in changes | Not estimated | Not estimated | 2.13 (-0.04 to 4.30) | Not estimated | Estimated |
| 65 to 70 | No documented hypercapnia | 20.28 (17.23 to 23.72) | 16.99 (13.72 to 20.84) | -3.29 (-3.67 to -2.92) | 0.84 (0.80 to 0.88) | Estimated |
| 65 to 70 | Hypercapnia without documented same-day acidemia | 21.30 (17.85 to 25.21) | 21.97 (17.59 to 27.08) | 0.67 (-0.46 to 1.80) | 1.03 (0.98 to 1.08) | Estimated |
| 65 to 70 | Hypercapnia with same-day acidemia | 58.42 (54.07 to 62.64) | 61.04 (55.59 to 66.24) | 2.62 (1.45 to 3.79) | 1.04 (1.03 to 1.06) | Estimated |
| 65 to 70 | Any hypercapnia | 79.72 (76.28 to 82.77) | 83.01 (79.16 to 86.28) | 3.29 (2.92 to 3.67) | 1.04 (1.04 to 1.04) | Estimated |
| 65 to 70 | With minus without: difference in changes | Not estimated | Not estimated | 1.95 (-0.31 to 4.22) | Not estimated | Estimated |
| 70 to 75 | No documented hypercapnia | 16.99 (13.72 to 20.84) | 14.13 (10.83 to 18.24) | -2.86 (-3.06 to -2.66) | 0.83 (0.79 to 0.88) | Estimated; sparse endpoint |
| 70 to 75 | Hypercapnia without documented same-day acidemia | 21.97 (17.59 to 27.08) | 22.51 (17.18 to 28.91) | 0.54 (-0.62 to 1.70) | 1.02 (0.98 to 1.07) | Estimated; sparse endpoint |
| 70 to 75 | Hypercapnia with same-day acidemia | 61.04 (55.59 to 66.24) | 63.36 (56.75 to 69.51) | 2.32 (1.13 to 3.50) | 1.04 (1.02 to 1.05) | Estimated; sparse endpoint |
| 70 to 75 | Any hypercapnia | 83.01 (79.16 to 86.28) | 85.87 (81.76 to 89.17) | 2.86 (2.66 to 3.06) | 1.03 (1.03 to 1.04) | Estimated; sparse endpoint |
| 70 to 75 | With minus without: difference in changes | Not estimated | Not estimated | 1.78 (-0.56 to 4.11) | Not estimated | Estimated; sparse endpoint |
| 75 to 80 | No documented hypercapnia | 14.13 (10.83 to 18.24) | 11.69 (8.48 to 15.90) | -2.45 (-2.55 to -2.34) | 0.83 (0.78 to 0.87) | Estimated; sparse endpoint |
| 75 to 80 | Hypercapnia without documented same-day acidemia | 22.51 (17.18 to 28.91) | 22.93 (16.67 to 30.67) | 0.42 (-0.76 to 1.60) | 1.02 (0.97 to 1.07) | Estimated; sparse endpoint |
| 75 to 80 | Hypercapnia with same-day acidemia | 63.36 (56.75 to 69.51) | 65.39 (57.57 to 72.45) | 2.03 (0.83 to 3.22) | 1.03 (1.02 to 1.05) | Estimated; sparse endpoint |

| BMI interval | Outcome | Lower BMI probability % (95% CI) | Higher BMI probability % (95% CI) | Change pp (95% CI) | Probability ratio (95% CI) | Status |
|---|---|---|---|---|---|---|
| 75 to 80 | Any hypercapnia | 85.87 (81.76 to 89.17) | 88.31 (84.10 to 91.52) | 2.45 (2.34 to 2.55) | 1.03 (1.03 to 1.03) | Estimated; sparse endpoint |
| 75 to 80 | With minus without: difference in changes | Not estimated | Not estimated | 1.61 (-0.77 to 3.99) | Not estimated | Estimated; sparse endpoint |
| 80 to 85 | No documented hypercapnia | 11.69 (8.48 to 15.90) | 9.61 (6.60 to 13.81) | -2.07 (-2.22 to -1.93) | 0.82 (0.78 to 0.87) | Estimated; sparse endpoint |
| 80 to 85 | Hypercapnia without documented same-day acidemia | 22.93 (16.67 to 30.67) | 23.24 (16.07 to 32.38) | 0.31 (-0.88 to 1.51) | 1.01 (0.97 to 1.06) | Estimated; sparse endpoint |
| 80 to 85 | Hypercapnia with same-day acidemia | 65.39 (57.57 to 72.45) | 67.15 (58.08 to 75.10) | 1.76 (0.55 to 2.97) | 1.03 (1.01 to 1.04) | Estimated; sparse endpoint |
| 80 to 85 | Any hypercapnia | 88.31 (84.10 to 91.52) | 90.39 (86.19 to 93.40) | 2.07 (1.93 to 2.22) | 1.02 (1.02 to 1.03) | Estimated; sparse endpoint |
| 80 to 85 | With minus without: difference in changes | Not estimated | Not estimated | 1.45 (-0.95 to 3.84) | Not estimated | Estimated; sparse endpoint |

Table 39: Joint probability and stacked-component reconciliation

| Field | Value |
|---|---|
| model id | primary::same day abg/encounter/multinomial spline |
| n grid | 225 |
| maximum probability sum error | 1.187939e-14 |
| maximum component sum error | 1.110223e-16 |
| status | assessed |
| analysis population id | primary |
| population id | primary |
| outcome id | three category |
| ascertainment window | encounter |
| model form | multinomial spline |

Requiring both analytes on the same date introduces additional selection. The reconciliation separates patient restriction, restriction to co-observed days, and addition of low pH. Values can represent different samples or periods before and after treatment; co-occurrence does not establish simultaneous hypercapnic acidemia, acute respiratory failure or chronic compensation. Primary arterial findings on other dates remain unchanged. The joint model guarantees coherent category probabilities without clipping. In the stacked panel, band thickness is the component probability and the upper boundary is total hypercapnia. The common-baseline panel shows both components with their own intervals. The signed with-minus-without change contrasts use joint covariance; interval overlap is not the comparison test. Neither documented category establishes chronic compensation or the mechanism of acidemia.

## 3.5 Documented respiratory-failure acuity and OHS

This supplement partitions the inherited five-code ICD composite on the same fixed primary indexes. It requires complete encounter diagnosis capture and nonmissing inherited adjustment variables, including retained Unknown categories; it does not require blood gas evidence. J96.22 or J96.02 together with J96.12/E66.2 receives priority as **acute-on-chronic or concurrent acute/chronic coding**. Remaining J96.02 is acute; remaining J96.12/E66.2 is chronic coding/OHS; remaining J96.92 is unspecified acuity. No target code with complete capture is the reference category. Incomplete capture is unknown. OHS is grouped analytically with chronic coding and does not itself establish chronic respiratory failure.

One Firth mean-bias-reduced five-category multinomial spline reuses the primary numeric BMI basis, linear age and inherited sex, race/ethnicity and region adjustment. All class probabilities are standardized over the same diagnosis-evaluable patients. The estimator maximizes the multinomial log likelihood plus one half of the log

determinant of its Fisher information. It replaces a separated unpenalized fit without pooling demographic levels or changing categories. The inverse multinomial information at the bias-reduced estimate supplies first-order joint covariance, with the empirical reference and numeric spline held fixed. This finite-estimate method does not resolve sparse support, source selection or measurement bias. Full joint covariance supplies pointwise 95% intervals, supported five-unit changes and pairwise differences in those changes. These descriptive comparisons are not multiplicity adjusted. The fitted positive components sum to the joint composite; it need not equal the existing binary model's predictions. Primary estimates, S1 and PIFs remain unchanged; no supplemental PIF is calculated.

Table 40: ICD code-level source qualification

| module | status | reason | n primary complete | n complete capture | analysis population id |
|---|---|---|---|---|---|
| icd acuity v1 | qualified | Complete encounter diagnosis capture; inherited five-code composite | 176922 | 176922 | primary |

Table 41: ICD model estimability; source qualification does not imply a successful fit

| model id | estimator | n model | status | failure reason |
|---|---|---|---|---|
| primary::icd acuity/ encounter/multinomial spline | firth mean bias reduced v1 | 176922 | successful | |
| primary::icd acuity specified/encounter/ multinomial spline | maximum likelihood | 176794 | successful | |

Table 42: Mutually exclusive ICD categories; observed proportions with Wilson 95% intervals

| Category | Patients | N | Observed | Status |
|---|---|---|---|---|
| No target diagnosis documented | 168288 | 176922 | 95.12% (95% CI 95.02–95.22%) | qualified |
| Chronic coding / OHS | 1769 | 176922 | 1.00% (95% CI 0.95–1.05%) | qualified |
| Acute-on-chronic or concurrent coding | 3443 | 176922 | 1.95% (95% CI 1.88–2.01%) | qualified |
| Acute coding | 3294 | 176922 | 1.86% (95% CI 1.80–1.93%) | qualified |
| Unspecified acuity | 128 | 176922 | 0.07% (95% CI 0.06–0.09%) | qualified |

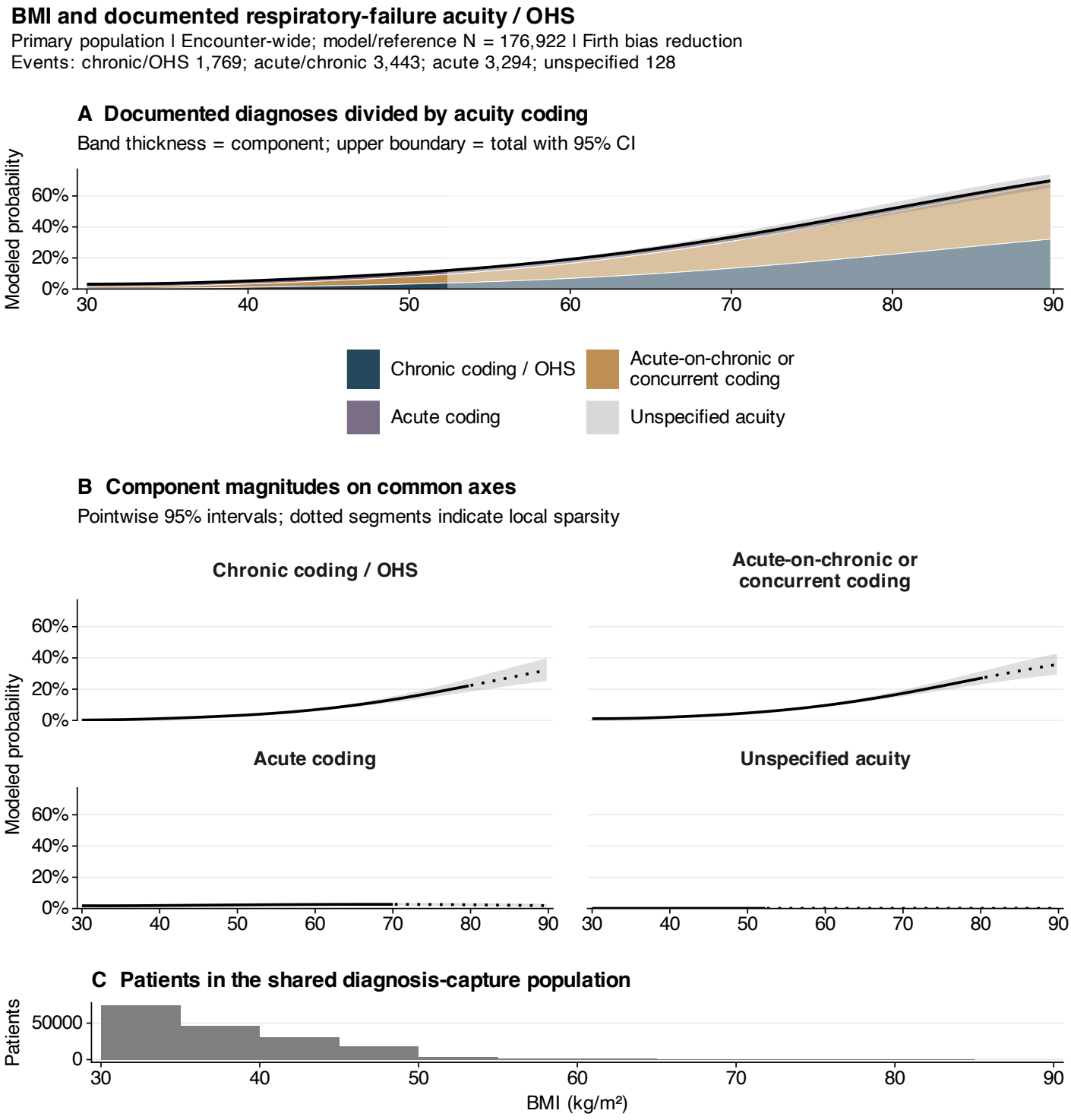



Figure 11: Mean-bias-reduced five-category multinomial spline; all diagnosis-evaluable primary patients retained. Encounter-wide coding, no gas requirement; inherited adjustment and primary spline basis. Bands partition the composite; thickness gives component probability. Shading: pointwise 95% intervals from joint information covariance, conditional on the empirical reference and basis. Fading/dots: local sparsity; histogram N equals model/reference N. Acute-on-chronic includes explicit or concurrent acute/chronic coding. OHS (obesity hypoventilation syndrome) is analytically chronic; this does not establish chronic respiratory failure. Source selection and diagnostic incorporation limit physiological interpretation.

### 3.5.1 Five-category bias-reduced model

Table 43: ICD component magnitudes and differences in BMI-associated changes; 95% intervals

| BMI interval | Outcome | Lower BMI probability % (95% CI) | Higher BMI probability % (95% CI) | Change pp (95% CI) | Probability ratio (95% CI) | Status |
|---|---|---|---|---|---|---|
| 30 to 35 | No target diagnosis documented | 96.88 (96.66 to 97.09) | 96.34 (96.19 to 96.49) | -0.55 (-0.84 to -0.25) | 0.99 (0.99 to 1.00) | Estimated |
| 30 to 35 | Chronic coding / OHS | 0.22 (0.17 to 0.28) | 0.51 (0.45 to 0.57) | 0.29 (0.20 to 0.37) | 2.29 (1.73 to 3.04) | Estimated |
| 30 to 35 | Acute-on-chronic or concurrent coding | 1.08 (0.96 to 1.21) | 1.34 (1.25 to 1.43) | 0.26 (0.09 to 0.43) | 1.24 (1.07 to 1.44) | Estimated |
| 30 to 35 | Acute coding | 1.75 (1.59 to 1.92) | 1.75 (1.64 to 1.86) | -0.00 (-0.23 to 0.23) | 1.00 (0.88 to 1.14) | Estimated |
| 30 to 35 | Unspecified acuity | 0.07 (0.04 to 0.11) | 0.07 (0.05 to 0.10) | 0.00 (-0.04 to 0.05) | 1.02 (0.53 to 1.98) | Estimated |
| 30 to 35 | Any target diagnosis | 3.12 (2.91 to 3.34) | 3.66 (3.51 to 3.81) | 0.55 (0.25 to 0.84) | 1.18 (1.07 to 1.29) | Estimated |
| 30 to 35 | Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -0.03 (-0.22 to 0.17) | Not estimated | Estimated |
| 30 to 35 | Acute coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -0.29 (-0.53 to -0.04) | Not estimated | Estimated |
| 30 to 35 | Unspecified acuity minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -0.28 (-0.38 to -0.19) | Not estimated | Estimated |
| 30 to 35 | Acute coding minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -0.26 (-0.55 to 0.03) | Not estimated | Estimated |
| 30 to 35 | Unspecified acuity minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -0.26 (-0.44 to -0.08) | Not estimated | Estimated |
| 30 to 35 | Unspecified acuity minus Acute coding : difference in changes | Not estimated | Not estimated | 0.00 (-0.23 to 0.23) | Not estimated | Estimated |
| 35 to 40 | No target diagnosis documented | 96.34 (96.19 to 96.49) | 94.77 (94.61 to 94.92) | -1.57 (-1.76 to -1.38) | 0.98 (0.98 to 0.99) | Estimated |
| 35 to 40 | Chronic coding / OHS | 0.51 (0.45 to 0.57) | 1.14 (1.07 to 1.22) | 0.64 (0.57 to 0.71) | 2.26 (2.04 to 2.49) | Estimated |
| 35 to 40 | Acute-on-chronic or concurrent coding | 1.34 (1.25 to 1.43) | 2.11 (2.01 to 2.21) | 0.77 (0.66 to 0.88) | 1.58 (1.47 to 1.69) | Estimated |
| 35 to 40 | Acute coding | 1.75 (1.64 to 1.86) | 1.90 (1.80 to 2.01) | 0.15 (0.02 to 0.29) | 1.09 (1.01 to 1.17) | Estimated |
| 35 to 40 | Unspecified acuity | 0.07 (0.05 to 0.10) | 0.08 (0.06 to 0.10) | 0.01 (-0.02 to 0.04) | 1.14 (0.79 to 1.65) | Estimated |
| 35 to 40 | Any target diagnosis | 3.66 (3.51 to 3.81) | 5.23 (5.08 to 5.39) | 1.57 (1.38 to 1.76) | 1.43 (1.37 to 1.49) | Estimated |
| 35 to 40 | Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | 0.13 (0.00 to 0.26) | Not estimated | Estimated |
| 35 to 40 | Acute coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -0.48 (-0.63 to -0.33) | Not estimated | Estimated |
| 35 to 40 | Unspecified acuity minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -0.63 (-0.70 to -0.55) | Not estimated | Estimated |
| 35 to 40 | Acute coding minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -0.62 (-0.79 to -0.44) | Not estimated | Estimated |

| BMI interval | Outcome | Lower BMI probability % (95% CI) | Higher BMI probability % (95% CI) | Change pp (95% CI) | Probability ratio (95% CI) | Status |
|---|---|---|---|---|---|---|
| 35 to 40 | Unspecified acuity minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -0.76 (-0.88 to -0.65) | Not estimated | Estimated |
| 35 to 40 | Unspecified acuity minus Acute coding : difference in changes | Not estimated | Not estimated | -0.14 (-0.28 to -0.01) | Not estimated | Estimated |
| 40 to 45 | No target diagnosis documented | 94.77 (94.61 to 94.92) | 92.54 (92.27 to 92.80) | -2.23 (-2.39 to -2.06) | 0.98 (0.97 to 0.98) | Estimated |
| 40 to 45 | Chronic coding / OHS | 1.14 (1.07 to 1.22) | 2.04 (1.90 to 2.19) | 0.90 (0.79 to 1.00) | 1.79 (1.70 to 1.88) | Estimated |
| 40 to 45 | Acute-on-chronic or concurrent coding | 2.11 (2.01 to 2.21) | 3.24 (3.06 to 3.43) | 1.13 (1.01 to 1.25) | 1.54 (1.49 to 1.59) | Estimated |
| 40 to 45 | Acute coding | 1.90 (1.80 to 2.01) | 2.09 (1.96 to 2.23) | 0.19 (0.11 to 0.26) | 1.10 (1.06 to 1.14) | Estimated |
| 40 to 45 | Unspecified acuity | 0.08 (0.06 to 0.10) | 0.09 (0.07 to 0.12) | 0.01 (-0.01 to 0.03) | 1.13 (0.95 to 1.35) | Estimated |
| 40 to 45 | Any target diagnosis | 5.23 (5.08 to 5.39) | 7.46 (7.20 to 7.73) | 2.23 (2.06 to 2.39) | 1.43 (1.40 to 1.46) | Estimated |
| 40 to 45 | Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | 0.23 (0.07 to 0.39) | Not estimated | Estimated |
| 40 to 45 | Acute coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -0.71 (-0.84 to -0.58) | Not estimated | Estimated |
| 40 to 45 | Unspecified acuity minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -0.89 (-1.00 to -0.78) | Not estimated | Estimated |
| 40 to 45 | Acute coding minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -0.94 (-1.08 to -0.80) | Not estimated | Estimated |
| 40 to 45 | Unspecified acuity minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -1.12 (-1.24 to -1.00) | Not estimated | Estimated |
| 40 to 45 | Unspecified acuity minus Acute coding : difference in changes | Not estimated | Not estimated | -0.18 (-0.25 to -0.10) | Not estimated | Estimated |
| 45 to 50 | No target diagnosis documented | 92.54 (92.27 to 92.80) | 89.71 (89.36 to 90.05) | -2.83 (-3.03 to -2.63) | 0.97 (0.97 to 0.97) | Estimated |
| 45 to 50 | Chronic coding / OHS | 2.04 (1.90 to 2.19) | 3.15 (2.96 to 3.37) | 1.11 (1.01 to 1.21) | 1.54 (1.49 to 1.60) | Estimated |
| 45 to 50 | Acute-on-chronic or concurrent coding | 3.24 (3.06 to 3.43) | 4.76 (4.52 to 5.01) | 1.52 (1.39 to 1.65) | 1.47 (1.43 to 1.51) | Estimated |
| 45 to 50 | Acute coding | 2.09 (1.96 to 2.23) | 2.28 (2.12 to 2.45) | 0.19 (0.08 to 0.30) | 1.09 (1.04 to 1.15) | Estimated |
| 45 to 50 | Unspecified acuity | 0.09 (0.07 to 0.12) | 0.10 (0.07 to 0.14) | 0.01 (-0.01 to 0.03) | 1.11 (0.87 to 1.41) | Estimated |
| 45 to 50 | Any target diagnosis | 7.46 (7.20 to 7.73) | 10.29 (9.95 to 10.64) | 2.83 (2.63 to 3.03) | 1.38 (1.35 to 1.41) | Estimated |
| 45 to 50 | Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | 0.41 (0.24 to 0.58) | Not estimated | Estimated |
| 45 to 50 | Acute coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -0.92 (-1.08 to -0.77) | Not estimated | Estimated |
| 45 to 50 | Unspecified acuity minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -1.10 (-1.21 to -0.99) | Not estimated | Estimated |

| BMI interval | Outcome | Lower BMI probability % (95% CI) | Higher BMI probability % (95% CI) | Change pp (95% CI) | Probability ratio (95% CI) | Status |
|---|---|---|---|---|---|---|
| 45 to 50 | Acute coding minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -1.33 (-1.50 to -1.15) | Not estimated | Estimated |
| 45 to 50 | Unspecified acuity minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -1.51 (-1.64 to -1.38) | Not estimated | Estimated |
| 45 to 50 | Unspecified acuity minus Acute coding : difference in changes | Not estimated | Not estimated | -0.18 (-0.29 to -0.06) | Not estimated | Estimated |
| 50 to 55 | No target diagnosis documented | 89.71 (89.36 to 90.05) | 85.90 (85.32 to 86.47) | -3.81 (-4.14 to -3.48) | 0.96 (0.95 to 0.96) | Estimated |
| 50 to 55 | Chronic coding / OHS | 3.15 (2.96 to 3.37) | 4.70 (4.36 to 5.05) | 1.54 (1.34 to 1.74) | 1.49 (1.43 to 1.55) | Estimated |
| 50 to 55 | Acute-on-chronic or concurrent coding | 4.76 (4.52 to 5.01) | 6.84 (6.43 to 7.27) | 2.08 (1.84 to 2.33) | 1.44 (1.39 to 1.48) | Estimated |
| 50 to 55 | Acute coding | 2.28 (2.12 to 2.45) | 2.45 (2.20 to 2.73) | 0.17 (0.03 to 0.31) | 1.08 (1.02 to 1.14) | Estimated |
| 50 to 55 | Unspecified acuity | 0.10 (0.07 to 0.14) | 0.11 (0.07 to 0.18) | 0.01 (-0.02 to 0.04) | 1.09 (0.84 to 1.43) | Estimated; sparse endpoint |
| 50 to 55 | Any target diagnosis | 10.29 (9.95 to 10.64) | 14.10 (13.53 to 14.68) | 3.81 (3.48 to 4.14) | 1.37 (1.34 to 1.40) | Estimated |
| 50 to 55 | Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | 0.54 (0.22 to 0.86) | Not estimated | Estimated |
| 50 to 55 | Acute coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -1.37 (-1.62 to -1.12) | Not estimated | Estimated |
| 50 to 55 | Unspecified acuity minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -1.53 (-1.74 to -1.33) | Not estimated | Estimated; sparse endpoint |
| 50 to 55 | Acute coding minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -1.91 (-2.20 to -1.62) | Not estimated | Estimated |
| 50 to 55 | Unspecified acuity minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -2.07 (-2.32 to -1.83) | Not estimated | Estimated; sparse endpoint |
| 50 to 55 | Unspecified acuity minus Acute coding : difference in changes | Not estimated | Not estimated | -0.16 (-0.31 to -0.02) | Not estimated | Estimated; sparse endpoint |
| 55 to 60 | No target diagnosis documented | 85.90 (85.32 to 86.47) | 80.84 (79.82 to 81.82) | -5.06 (-5.55 to -4.58) | 0.94 (0.94 to 0.95) | Estimated |
| 55 to 60 | Chronic coding / OHS | 4.70 (4.36 to 5.05) | 6.85 (6.22 to 7.53) | 2.15 (1.80 to 2.49) | 1.46 (1.41 to 1.51) | Estimated |
| 55 to 60 | Acute-on-chronic or concurrent coding | 6.84 (6.43 to 7.27) | 9.62 (8.88 to 10.41) | 2.78 (2.38 to 3.17) | 1.41 (1.37 to 1.45) | Estimated |
| 55 to 60 | Acute coding | 2.45 (2.20 to 2.73) | 2.58 (2.21 to 3.02) | 0.13 (-0.02 to 0.29) | 1.05 (1.00 to 1.11) | Estimated |
| 55 to 60 | Unspecified acuity | 0.11 (0.07 to 0.18) | 0.12 (0.06 to 0.25) | 0.01 (-0.03 to 0.04) | 1.07 (0.82 to 1.40) | Estimated; sparse endpoint |
| 55 to 60 | Any target diagnosis | 14.10 (13.53 to 14.68) | 19.16 (18.18 to 20.18) | 5.06 (4.58 to 5.55) | 1.36 (1.33 to 1.38) | Estimated |
| 55 to 60 | Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | 0.63 (0.07 to 1.19) | Not estimated | Estimated |

| BMI interval | Outcome | Lower BMI probability % (95% CI) | Higher BMI probability % (95% CI) | Change pp (95% CI) | Probability ratio (95% CI) | Status |
|---|---|---|---|---|---|---|
| 55 to 60 | Acute coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -2.02 (-2.41 to -1.63) | Not estimated | Estimated |
| 55 to 60 | Unspecified acuity minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -2.14 (-2.49 to -1.79) | Not estimated | Estimated; sparse endpoint |
| 55 to 60 | Acute coding minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -2.64 (-3.08 to -2.20) | Not estimated | Estimated |
| 55 to 60 | Unspecified acuity minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -2.77 (-3.17 to -2.37) | Not estimated | Estimated; sparse endpoint |
| 55 to 60 | Unspecified acuity minus Acute coding : difference in changes | Not estimated | Not estimated | -0.12 (-0.28 to 0.04) | Not estimated | Estimated; sparse endpoint |
| 60 to 65 | No target diagnosis documented | 80.84 (79.82 to 81.82) | 74.38 (72.72 to 75.96) | -6.46 (-7.13 to -5.80) | 0.92 (0.91 to 0.93) | Estimated |
| 60 to 65 | Chronic coding / OHS | 6.85 (6.22 to 7.53) | 9.71 (8.60 to 10.94) | 2.86 (2.31 to 3.41) | 1.42 (1.37 to 1.47) | Estimated |
| 60 to 65 | Acute-on-chronic or concurrent coding | 9.62 (8.88 to 10.41) | 13.14 (11.88 to 14.52) | 3.53 (2.94 to 4.11) | 1.37 (1.33 to 1.40) | Estimated |
| 60 to 65 | Acute coding | 2.58 (2.21 to 3.02) | 2.65 (2.15 to 3.27) | 0.07 (-0.09 to 0.23) | 1.03 (0.97 to 1.09) | Estimated |
| 60 to 65 | Unspecified acuity | 0.12 (0.06 to 0.25) | 0.12 (0.05 to 0.33) | 0.01 (-0.03 to 0.04) | 1.04 (0.80 to 1.36) | Estimated; sparse endpoint |
| 60 to 65 | Any target diagnosis | 19.16 (18.18 to 20.18) | 25.62 (24.04 to 27.28) | 6.46 (5.80 to 7.13) | 1.34 (1.32 to 1.36) | Estimated |
| 60 to 65 | Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | 0.66 (-0.24 to 1.57) | Not estimated | Estimated |
| 60 to 65 | Acute coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -2.79 (-3.38 to -2.20) | Not estimated | Estimated |
| 60 to 65 | Unspecified acuity minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -2.86 (-3.41 to -2.30) | Not estimated | Estimated; sparse endpoint |
| 60 to 65 | Acute coding minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -3.45 (-4.08 to -2.83) | Not estimated | Estimated |
| 60 to 65 | Unspecified acuity minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -3.52 (-4.11 to -2.93) | Not estimated | Estimated; sparse endpoint |
| 60 to 65 | Unspecified acuity minus Acute coding : difference in changes | Not estimated | Not estimated | -0.07 (-0.23 to 0.10) | Not estimated | Estimated; sparse endpoint |
| 65 to 70 | No target diagnosis documented | 74.38 (72.72 to 75.96) | 66.55 (64.13 to 68.89) | -7.83 (-8.62 to -7.03) | 0.89 (0.88 to 0.91) | Estimated |
| 65 to 70 | Chronic coding / OHS | 9.71 (8.60 to 10.94) | 13.32 (11.49 to 15.39) | 3.61 (2.82 to 4.41) | 1.37 (1.33 to 1.41) | Estimated |
| 65 to 70 | Acute-on-chronic or concurrent coding | 13.14 (11.88 to 14.52) | 17.36 (15.37 to 19.55) | 4.22 (3.43 to 5.01) | 1.32 (1.29 to 1.35) | Estimated |
| 65 to 70 | Acute coding | 2.65 (2.15 to 3.27) | 2.64 (2.03 to 3.43) | -0.01 (-0.16 to 0.14) | 1.00 (0.94 to 1.05) | Estimated |
| 65 to 70 | Unspecified acuity | 0.12 (0.05 to 0.33) | 0.12 (0.04 to 0.43) | 0.00 (-0.03 to 0.04) | 1.01 (0.77 to 1.32) | Estimated; sparse endpoint |
| 65 to 70 | Any target diagnosis | 25.62 (24.04 to 27.28) | 33.45 (31.11 to 35.87) | 7.83 (7.03 to 8.62) | 1.31 (1.29 to 1.32) | Estimated |

| BMI interval | Outcome | Lower BMI probability % (95% CI) | Higher BMI probability % (95% CI) | Change pp (95% CI) | Probability ratio (95% CI) | Status |
|---|---|---|---|---|---|---|
| 65 to 70 | Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | 0.61 (-0.74 to 1.96) | Not estimated | Estimated |
| 65 to 70 | Acute coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -3.62 (-4.46 to -2.79) | Not estimated | Estimated |
| 65 to 70 | Unspecified acuity minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -3.61 (-4.41 to -2.81) | Not estimated | Estimated; sparse endpoint |
| 65 to 70 | Acute coding minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -4.23 (-5.06 to -3.40) | Not estimated | Estimated |
| 65 to 70 | Unspecified acuity minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -4.22 (-5.01 to -3.43) | Not estimated | Estimated; sparse endpoint |
| 65 to 70 | Unspecified acuity minus Acute coding : difference in changes | Not estimated | Not estimated | 0.01 (-0.14 to 0.17) | Not estimated | Estimated; sparse endpoint |
| 70 to 75 | No target diagnosis documented | 66.55 (64.13 to 68.89) | 57.64 (54.44 to 60.77) | -8.92 (-9.74 to -8.09) | 0.87 (0.85 to 0.88) | Estimated |
| 70 to 75 | Chronic coding / OHS | 13.32 (11.49 to 15.39) | 17.62 (14.83 to 20.80) | 4.30 (3.23 to 5.36) | 1.32 (1.29 to 1.36) | Estimated |
| 70 to 75 | Acute-on-chronic or concurrent coding | 17.36 (15.37 to 19.55) | 22.09 (19.19 to 25.29) | 4.72 (3.72 to 5.72) | 1.27 (1.24 to 1.30) | Estimated |
| 70 to 75 | Acute coding | 2.64 (2.03 to 3.43) | 2.54 (1.84 to 3.49) | -0.10 (-0.23 to 0.02) | 0.96 (0.91 to 1.02) | Estimated; sparse endpoint |
| 70 to 75 | Unspecified acuity | 0.12 (0.04 to 0.43) | 0.12 (0.03 to 0.55) | -0.00 (-0.03 to 0.03) | 0.98 (0.75 to 1.28) | Estimated; sparse endpoint |
| 70 to 75 | Any target diagnosis | 33.45 (31.11 to 35.87) | 42.36 (39.23 to 45.56) | 8.92 (8.09 to 9.74) | 1.27 (1.26 to 1.28) | Estimated |
| 70 to 75 | Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | 0.43 (-1.44 to 2.30) | Not estimated | Estimated |
| 70 to 75 | Acute coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -4.40 (-5.50 to -3.30) | Not estimated | Estimated; sparse endpoint |
| 70 to 75 | Unspecified acuity minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -4.30 (-5.37 to -3.23) | Not estimated | Estimated; sparse endpoint |
| 70 to 75 | Acute coding minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -4.83 (-5.86 to -3.80) | Not estimated | Estimated; sparse endpoint |
| 70 to 75 | Unspecified acuity minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -4.73 (-5.73 to -3.73) | Not estimated | Estimated; sparse endpoint |
| 70 to 75 | Unspecified acuity minus Acute coding : difference in changes | Not estimated | Not estimated | 0.10 (-0.03 to 0.23) | Not estimated | Estimated; sparse endpoint |
| 75 to 80 | No target diagnosis documented | 57.64 (54.44 to 60.77) | 48.13 (44.30 to 51.98) | -9.50 (-10.22 to -8.79) | 0.84 (0.81 to 0.86) | Estimated |
| 75 to 80 | Chronic coding / OHS | 17.62 (14.83 to 20.80) | 22.41 (18.43 to 26.97) | 4.79 (3.47 to 6.11) | 1.27 (1.24 to 1.31) | Estimated; sparse endpoint |

| BMI interval | Outcome | Lower BMI probability % (95% CI) | Higher BMI probability % (95% CI) | Change pp (95% CI) | Probability ratio (95% CI) | Status |
|---|---|---|---|---|---|---|
| 75 to 80 | Acute-on-chronic or concurrent coding | 22.09 (19.19 to 25.29) | 27.00 (23.02 to 31.38) | 4.91 (3.71 to 6.11) | 1.22 (1.19 to 1.25) | Estimated |
| 75 to 80 | Acute coding | 2.54 (1.84 to 3.49) | 2.35 (1.61 to 3.42) | -0.19 (-0.28 to -0.10) | 0.93 (0.87 to 0.98) | Estimated; sparse endpoint |
| 75 to 80 | Unspecified acuity | 0.12 (0.03 to 0.55) | 0.11 (0.02 to 0.67) | -0.01 (-0.03 to 0.01) | 0.94 (0.72 to 1.23) | Estimated; sparse endpoint |
| 75 to 80 | Any target diagnosis | 42.36 (39.23 to 45.56) | 51.87 (48.02 to 55.70) | 9.50 (8.79 to 10.22) | 1.22 (1.22 to 1.23) | Estimated |
| 75 to 80 | Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | 0.12 (-2.28 to 2.52) | Not estimated | Estimated; sparse endpoint |
| 75 to 80 | Acute coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -4.98 (-6.32 to -3.64) | Not estimated | Estimated; sparse endpoint |
| 75 to 80 | Unspecified acuity minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -4.80 (-6.12 to -3.47) | Not estimated | Estimated; sparse endpoint |
| 75 to 80 | Acute coding minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -5.10 (-6.32 to -3.89) | Not estimated | Estimated; sparse endpoint |
| 75 to 80 | Unspecified acuity minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -4.92 (-6.12 to -3.72) | Not estimated | Estimated; sparse endpoint |
| 75 to 80 | Unspecified acuity minus Acute coding : difference in changes | Not estimated | Not estimated | 0.18 (0.09 to 0.27) | Not estimated | Estimated; sparse endpoint |
| 80 to 85 | No target diagnosis documented | 48.13 (44.30 to 51.98) | 38.68 (34.51 to 43.02) | -9.45 (-9.92 to -8.98) | 0.80 (0.78 to 0.83) | Estimated; sparse endpoint |
| 80 to 85 | Chronic coding / OHS | 22.41 (18.43 to 26.97) | 27.40 (22.04 to 33.51) | 5.00 (3.47 to 6.53) | 1.22 (1.19 to 1.25) | Estimated; sparse endpoint |
| 80 to 85 | Acute-on-chronic or concurrent coding | 27.00 (23.02 to 31.38) | 31.72 (26.51 to 37.43) | 4.72 (3.34 to 6.11) | 1.17 (1.15 to 1.20) | Estimated; sparse endpoint |
| 80 to 85 | Acute coding | 2.35 (1.61 to 3.42) | 2.09 (1.34 to 3.23) | -0.26 (-0.31 to -0.21) | 0.89 (0.83 to 0.95) | Estimated; sparse endpoint |
| 80 to 85 | Unspecified acuity | 0.11 (0.02 to 0.67) | 0.10 (0.01 to 0.79) | -0.01 (-0.02 to -0.00) | 0.90 (0.69 to 1.18) | Estimated; sparse endpoint |
| 80 to 85 | Any target diagnosis | 51.87 (48.02 to 55.70) | 61.32 (56.98 to 65.49) | 9.45 (8.98 to 9.92) | 1.18 (1.18 to 1.19) | Estimated; sparse endpoint |
| 80 to 85 | Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -0.27 (-3.14 to 2.60) | Not estimated | Estimated; sparse endpoint |
| 80 to 85 | Acute coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -5.26 (-6.80 to -3.71) | Not estimated | Estimated; sparse endpoint |
| 80 to 85 | Unspecified acuity minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -5.01 (-6.54 to -3.48) | Not estimated | Estimated; sparse endpoint |
| 80 to 85 | Acute coding minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -4.98 (-6.37 to -3.60) | Not estimated | Estimated; sparse endpoint |

| BMI interval | Outcome | Lower BMI probability % (95% CI) | Higher BMI probability % (95% CI) | Change pp (95% CI) | Probability ratio (95% CI) | Status |
|---|---|---|---|---|---|---|
| 80 to 85 | Unspecified acuity minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -4.73 (-6.12 to -3.35) | Not estimated | Estimated; sparse endpoint |
| 80 to 85 | Unspecified acuity minus Acute coding : difference in changes | Not estimated | Not estimated | 0.25 (0.20 to 0.30) | Not estimated | Estimated; sparse endpoint |

### 3.5.2 Four-category sensitivity

Table 44: ICD component magnitudes and differences in BMI-associated changes; 95% intervals

| BMI interval | Outcome | Lower BMI probability % (95% CI) | Higher BMI probability % (95% CI) | Change pp (95% CI) | Probability ratio (95% CI) | Status |
|---|---|---|---|---|---|---|
| 30 to 35 | No target diagnosis documented | 96.96 (96.74 to 97.17) | 96.41 (96.26 to 96.56) | -0.54 (-0.84 to -0.25) | 0.99 (0.99 to 1.00) | Estimated |
| 30 to 35 | Chronic coding / OHS | 0.22 (0.17 to 0.28) | 0.51 (0.45 to 0.56) | 0.29 (0.20 to 0.37) | 2.30 (1.73 to 3.05) | Estimated |
| 30 to 35 | Acute-on-chronic or concurrent coding | 1.08 (0.96 to 1.21) | 1.34 (1.25 to 1.43) | 0.26 (0.09 to 0.43) | 1.24 (1.07 to 1.44) | Estimated |
| 30 to 35 | Acute coding | 1.74 (1.58 to 1.92) | 1.74 (1.64 to 1.85) | -0.00 (-0.23 to 0.23) | 1.00 (0.88 to 1.14) | Estimated |
| 30 to 35 | Any target diagnosis | 3.04 (2.83 to 3.26) | 3.59 (3.44 to 3.74) | 0.54 (0.25 to 0.84) | 1.18 (1.08 to 1.29) | Estimated |
| 30 to 35 | Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -0.03 (-0.22 to 0.17) | Not estimated | Estimated |
| 30 to 35 | Acute coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -0.29 (-0.53 to -0.04) | Not estimated | Estimated |
| 30 to 35 | Acute coding minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -0.26 (-0.55 to 0.03) | Not estimated | Estimated |
| 35 to 40 | No target diagnosis documented | 96.41 (96.26 to 96.56) | 94.85 (94.69 to 95.01) | -1.56 (-1.75 to -1.38) | 0.98 (0.98 to 0.99) | Estimated |
| 35 to 40 | Chronic coding / OHS | 0.51 (0.45 to 0.56) | 1.14 (1.07 to 1.21) | 0.64 (0.57 to 0.70) | 2.26 (2.04 to 2.49) | Estimated |
| 35 to 40 | Acute-on-chronic or concurrent coding | 1.34 (1.25 to 1.43) | 2.11 (2.01 to 2.21) | 0.77 (0.66 to 0.88) | 1.58 (1.47 to 1.69) | Estimated |
| 35 to 40 | Acute coding | 1.74 (1.64 to 1.85) | 1.90 (1.80 to 2.01) | 0.15 (0.02 to 0.29) | 1.09 (1.01 to 1.17) | Estimated |
| 35 to 40 | Any target diagnosis | 3.59 (3.44 to 3.74) | 5.15 (4.99 to 5.31) | 1.56 (1.38 to 1.75) | 1.44 (1.37 to 1.50) | Estimated |
| 35 to 40 | Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | 0.14 (0.00 to 0.27) | Not estimated | Estimated |
| 35 to 40 | Acute coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -0.48 (-0.63 to -0.33) | Not estimated | Estimated |
| 35 to 40 | Acute coding minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -0.62 (-0.79 to -0.44) | Not estimated | Estimated |
| 40 to 45 | No target diagnosis documented | 94.85 (94.69 to 95.01) | 92.64 (92.37 to 92.90) | -2.21 (-2.38 to -2.05) | 0.98 (0.97 to 0.98) | Estimated |
| 40 to 45 | Chronic coding / OHS | 1.14 (1.07 to 1.21) | 2.04 (1.90 to 2.19) | 0.90 (0.79 to 1.00) | 1.79 (1.70 to 1.88) | Estimated |

| BMI interval | Outcome | Lower BMI probability % (95% CI) | Higher BMI probability % (95% CI) | Change pp (95% CI) | Probability ratio (95% CI) | Status |
|---|---|---|---|---|---|---|
| 40 to 45 | Acute-on-chronic or concurrent coding | 2.11 (2.01 to 2.21) | 3.24 (3.06 to 3.42) | 1.13 (1.01 to 1.25) | 1.54 (1.49 to 1.59) | Estimated |
| 40 to 45 | Acute coding | 1.90 (1.80 to 2.01) | 2.09 (1.95 to 2.23) | 0.19 (0.11 to 0.26) | 1.10 (1.06 to 1.14) | Estimated |
| 40 to 45 | Any target diagnosis | 5.15 (4.99 to 5.31) | 7.36 (7.10 to 7.63) | 2.21 (2.05 to 2.38) | 1.43 (1.40 to 1.46) | Estimated |
| 40 to 45 | Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | 0.23 (0.07 to 0.39) | Not estimated | Estimated |
| 40 to 45 | Acute coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -0.71 (-0.84 to -0.58) | Not estimated | Estimated |
| 40 to 45 | Acute coding minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -0.94 (-1.08 to -0.80) | Not estimated | Estimated |
| 45 to 50 | No target diagnosis documented | 92.64 (92.37 to 92.90) | 89.82 (89.48 to 90.16) | -2.81 (-3.01 to -2.62) | 0.97 (0.97 to 0.97) | Estimated |
| 45 to 50 | Chronic coding / OHS | 2.04 (1.90 to 2.19) | 3.15 (2.95 to 3.36) | 1.11 (1.00 to 1.21) | 1.54 (1.49 to 1.60) | Estimated |
| 45 to 50 | Acute-on-chronic or concurrent coding | 3.24 (3.06 to 3.42) | 4.76 (4.51 to 5.01) | 1.52 (1.39 to 1.65) | 1.47 (1.43 to 1.51) | Estimated |
| 45 to 50 | Acute coding | 2.09 (1.95 to 2.23) | 2.27 (2.11 to 2.44) | 0.19 (0.07 to 0.30) | 1.09 (1.04 to 1.15) | Estimated |
| 45 to 50 | Any target diagnosis | 7.36 (7.10 to 7.63) | 10.18 (9.84 to 10.52) | 2.81 (2.62 to 3.01) | 1.38 (1.35 to 1.41) | Estimated |
| 45 to 50 | Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | 0.41 (0.24 to 0.58) | Not estimated | Estimated |
| 45 to 50 | Acute coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -0.92 (-1.08 to -0.77) | Not estimated | Estimated |
| 45 to 50 | Acute coding minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -1.33 (-1.51 to -1.16) | Not estimated | Estimated |
| 50 to 55 | No target diagnosis documented | 89.82 (89.48 to 90.16) | 86.03 (85.45 to 86.59) | -3.79 (-4.12 to -3.47) | 0.96 (0.95 to 0.96) | Estimated |
| 50 to 55 | Chronic coding / OHS | 3.15 (2.95 to 3.36) | 4.69 (4.35 to 5.04) | 1.54 (1.34 to 1.74) | 1.49 (1.43 to 1.55) | Estimated |
| 50 to 55 | Acute-on-chronic or concurrent coding | 4.76 (4.51 to 5.01) | 6.84 (6.43 to 7.27) | 2.08 (1.84 to 2.33) | 1.44 (1.39 to 1.48) | Estimated |
| 50 to 55 | Acute coding | 2.27 (2.11 to 2.44) | 2.44 (2.19 to 2.72) | 0.17 (0.03 to 0.31) | 1.07 (1.02 to 1.14) | Estimated |
| 50 to 55 | Any target diagnosis | 10.18 (9.84 to 10.52) | 13.97 (13.41 to 14.55) | 3.79 (3.47 to 4.12) | 1.37 (1.34 to 1.40) | Estimated |
| 50 to 55 | Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | 0.54 (0.22 to 0.86) | Not estimated | Estimated |
| 50 to 55 | Acute coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -1.37 (-1.62 to -1.12) | Not estimated | Estimated |
| 50 to 55 | Acute coding minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -1.91 (-2.20 to -1.62) | Not estimated | Estimated |
| 55 to 60 | No target diagnosis documented | 86.03 (85.45 to 86.59) | 80.98 (79.96 to 81.96) | -5.05 (-5.54 to -4.56) | 0.94 (0.94 to 0.95) | Estimated |
| 55 to 60 | Chronic coding / OHS | 4.69 (4.35 to 5.04) | 6.83 (6.21 to 7.51) | 2.14 (1.80 to 2.49) | 1.46 (1.41 to 1.51) | Estimated |
| 55 to 60 | Acute-on-chronic or concurrent coding | 6.84 (6.43 to 7.27) | 9.61 (8.87 to 10.41) | 2.78 (2.38 to 3.17) | 1.41 (1.37 to 1.45) | Estimated |

| BMI interval | Outcome | Lower BMI probability % (95% CI) | Higher BMI probability % (95% CI) | Change pp (95% CI) | Probability ratio (95% CI) | Status |
|---|---|---|---|---|---|---|
| 55 to 60 | Acute coding | 2.44 (2.19 to 2.72) | 2.57 (2.20 to 3.01) | 0.13 (-0.03 to 0.29) | 1.05 (1.00 to 1.11) | Estimated |
| 55 to 60 | Any target diagnosis | 13.97 (13.41 to 14.55) | 19.02 (18.04 to 20.04) | 5.05 (4.56 to 5.54) | 1.36 (1.34 to 1.39) | Estimated |
| 55 to 60 | Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | 0.63 (0.07 to 1.19) | Not estimated | Estimated |
| 55 to 60 | Acute coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -2.01 (-2.40 to -1.62) | Not estimated | Estimated |
| 55 to 60 | Acute coding minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -2.65 (-3.09 to -2.21) | Not estimated | Estimated |
| 60 to 65 | No target diagnosis documented | 80.98 (79.96 to 81.96) | 74.53 (72.87 to 76.11) | -6.45 (-7.12 to -5.79) | 0.92 (0.91 to 0.93) | Estimated |
| 60 to 65 | Chronic coding / OHS | 6.83 (6.21 to 7.51) | 9.69 (8.58 to 10.92) | 2.86 (2.31 to 3.41) | 1.42 (1.37 to 1.47) | Estimated |
| 60 to 65 | Acute-on-chronic or concurrent coding | 9.61 (8.87 to 10.41) | 13.14 (11.87 to 14.52) | 3.53 (2.94 to 4.11) | 1.37 (1.33 to 1.40) | Estimated |
| 60 to 65 | Acute coding | 2.57 (2.20 to 3.01) | 2.64 (2.14 to 3.26) | 0.07 (-0.09 to 0.23) | 1.03 (0.97 to 1.09) | Estimated |
| 60 to 65 | Any target diagnosis | 19.02 (18.04 to 20.04) | 25.47 (23.89 to 27.13) | 6.45 (5.79 to 7.12) | 1.34 (1.32 to 1.36) | Estimated |
| 60 to 65 | Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | 0.67 (-0.24 to 1.58) | Not estimated | Estimated |
| 60 to 65 | Acute coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -2.79 (-3.38 to -2.20) | Not estimated | Estimated |
| 60 to 65 | Acute coding minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -3.46 (-4.08 to -2.83) | Not estimated | Estimated |
| 65 to 70 | No target diagnosis documented | 74.53 (72.87 to 76.11) | 66.71 (64.28 to 69.05) | -7.82 (-8.62 to -7.02) | 0.90 (0.88 to 0.91) | Estimated |
| 65 to 70 | Chronic coding / OHS | 9.69 (8.58 to 10.92) | 13.30 (11.47 to 15.37) | 3.61 (2.81 to 4.41) | 1.37 (1.33 to 1.42) | Estimated |
| 65 to 70 | Acute-on-chronic or concurrent coding | 13.14 (11.87 to 14.52) | 17.36 (15.37 to 19.55) | 4.22 (3.43 to 5.02) | 1.32 (1.29 to 1.35) | Estimated |
| 65 to 70 | Acute coding | 2.64 (2.14 to 3.26) | 2.63 (2.02 to 3.42) | -0.01 (-0.16 to 0.14) | 1.00 (0.94 to 1.05) | Estimated |
| 65 to 70 | Any target diagnosis | 25.47 (23.89 to 27.13) | 33.29 (30.95 to 35.72) | 7.82 (7.02 to 8.62) | 1.31 (1.29 to 1.32) | Estimated |
| 65 to 70 | Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | 0.61 (-0.74 to 1.97) | Not estimated | Estimated |
| 65 to 70 | Acute coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -3.62 (-4.46 to -2.79) | Not estimated | Estimated |
| 65 to 70 | Acute coding minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -4.24 (-5.07 to -3.40) | Not estimated | Estimated |
| 70 to 75 | No target diagnosis documented | 66.71 (64.28 to 69.05) | 57.79 (54.58 to 60.93) | -8.92 (-9.75 to -8.09) | 0.87 (0.85 to 0.88) | Estimated |
| 70 to 75 | Chronic coding / OHS | 13.30 (11.47 to 15.37) | 17.59 (14.80 to 20.78) | 4.29 (3.23 to 5.36) | 1.32 (1.29 to 1.36) | Estimated |
| 70 to 75 | Acute-on-chronic or concurrent coding | 17.36 (15.37 to 19.55) | 22.09 (19.19 to 25.29) | 4.73 (3.73 to 5.73) | 1.27 (1.24 to 1.30) | Estimated |
| 70 to 75 | Acute coding | 2.63 (2.02 to 3.42) | 2.53 (1.83 to 3.48) | -0.10 (-0.23 to 0.02) | 0.96 (0.91 to 1.02) | Estimated; sparse endpoint |

| BMI interval | Outcome | Lower BMI probability % (95% CI) | Higher BMI probability % (95% CI) | Change pp (95% CI) | Probability ratio (95% CI) | Status |
|---|---|---|---|---|---|---|
| 70 to 75 | Any target diagnosis | 33.29 (30.95 to 35.72) | 42.21 (39.07 to 45.42) | 8.92 (8.09 to 9.75) | 1.27 (1.26 to 1.28) | Estimated |
| 70 to 75 | Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | 0.43 (-1.44 to 2.31) | Not estimated | Estimated |
| 70 to 75 | Acute coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -4.40 (-5.50 to -3.30) | Not estimated | Estimated; sparse endpoint |
| 70 to 75 | Acute coding minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -4.83 (-5.86 to -3.80) | Not estimated | Estimated; sparse endpoint |
| 75 to 80 | No target diagnosis documented | 57.79 (54.58 to 60.93) | 48.28 (44.44 to 52.14) | -9.51 (-10.23 to -8.80) | 0.84 (0.81 to 0.86) | Estimated |
| 75 to 80 | Chronic coding / OHS | 17.59 (14.80 to 20.78) | 22.38 (18.40 to 26.94) | 4.79 (3.47 to 6.11) | 1.27 (1.24 to 1.31) | Estimated; sparse endpoint |
| 75 to 80 | Acute-on-chronic or concurrent coding | 22.09 (19.19 to 25.29) | 27.01 (23.03 to 31.40) | 4.92 (3.72 to 6.11) | 1.22 (1.19 to 1.25) | Estimated |
| 75 to 80 | Acute coding | 2.53 (1.83 to 3.48) | 2.34 (1.60 to 3.41) | -0.19 (-0.28 to -0.10) | 0.92 (0.87 to 0.98) | Estimated; sparse endpoint |
| 75 to 80 | Any target diagnosis | 42.21 (39.07 to 45.42) | 51.72 (47.86 to 55.56) | 9.51 (8.80 to 10.23) | 1.23 (1.22 to 1.23) | Estimated |
| 75 to 80 | Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | 0.13 (-2.28 to 2.53) | Not estimated | Estimated; sparse endpoint |
| 75 to 80 | Acute coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -4.98 (-6.32 to -3.63) | Not estimated | Estimated; sparse endpoint |
| 75 to 80 | Acute coding minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -5.11 (-6.32 to -3.89) | Not estimated | Estimated; sparse endpoint |
| 80 to 85 | No target diagnosis documented | 48.28 (44.44 to 52.14) | 38.81 (34.63 to 43.16) | -9.47 (-9.94 to -8.99) | 0.80 (0.78 to 0.83) | Estimated; sparse endpoint |
| 80 to 85 | Chronic coding / OHS | 22.38 (18.40 to 26.94) | 27.38 (22.01 to 33.49) | 5.00 (3.47 to 6.53) | 1.22 (1.19 to 1.25) | Estimated; sparse endpoint |
| 80 to 85 | Acute-on-chronic or concurrent coding | 27.01 (23.03 to 31.40) | 31.74 (26.52 to 37.45) | 4.73 (3.34 to 6.11) | 1.18 (1.15 to 1.20) | Estimated; sparse endpoint |
| 80 to 85 | Acute coding | 2.34 (1.60 to 3.41) | 2.08 (1.34 to 3.22) | -0.26 (-0.31 to -0.21) | 0.89 (0.83 to 0.95) | Estimated; sparse endpoint |
| 80 to 85 | Any target diagnosis | 51.72 (47.86 to 55.56) | 61.19 (56.84 to 65.37) | 9.47 (8.99 to 9.94) | 1.18 (1.18 to 1.19) | Estimated; sparse endpoint |
| 80 to 85 | Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -0.27 (-3.14 to 2.60) | Not estimated | Estimated; sparse endpoint |
| 80 to 85 | Acute coding minus Chronic coding / OHS : difference in changes | Not estimated | Not estimated | -5.26 (-6.80 to -3.71) | Not estimated | Estimated; sparse endpoint |
| 80 to 85 | Acute coding minus Acute-on-chronic or concurrent coding : difference in changes | Not estimated | Not estimated | -4.99 (-6.37 to -3.60) | Not estimated | Estimated; sparse endpoint |

The unpenalized four-category sensitivity excludes only the final unspecified-only category. Mixed codes assigned

to specified categories remain included. It retains the primary basis and adjustment but uses its own reduced fitting and standardization population; this is not a same-denominator estimator comparison. Persistent separation is an explicit failure, not an invitation to substitute an estimator.

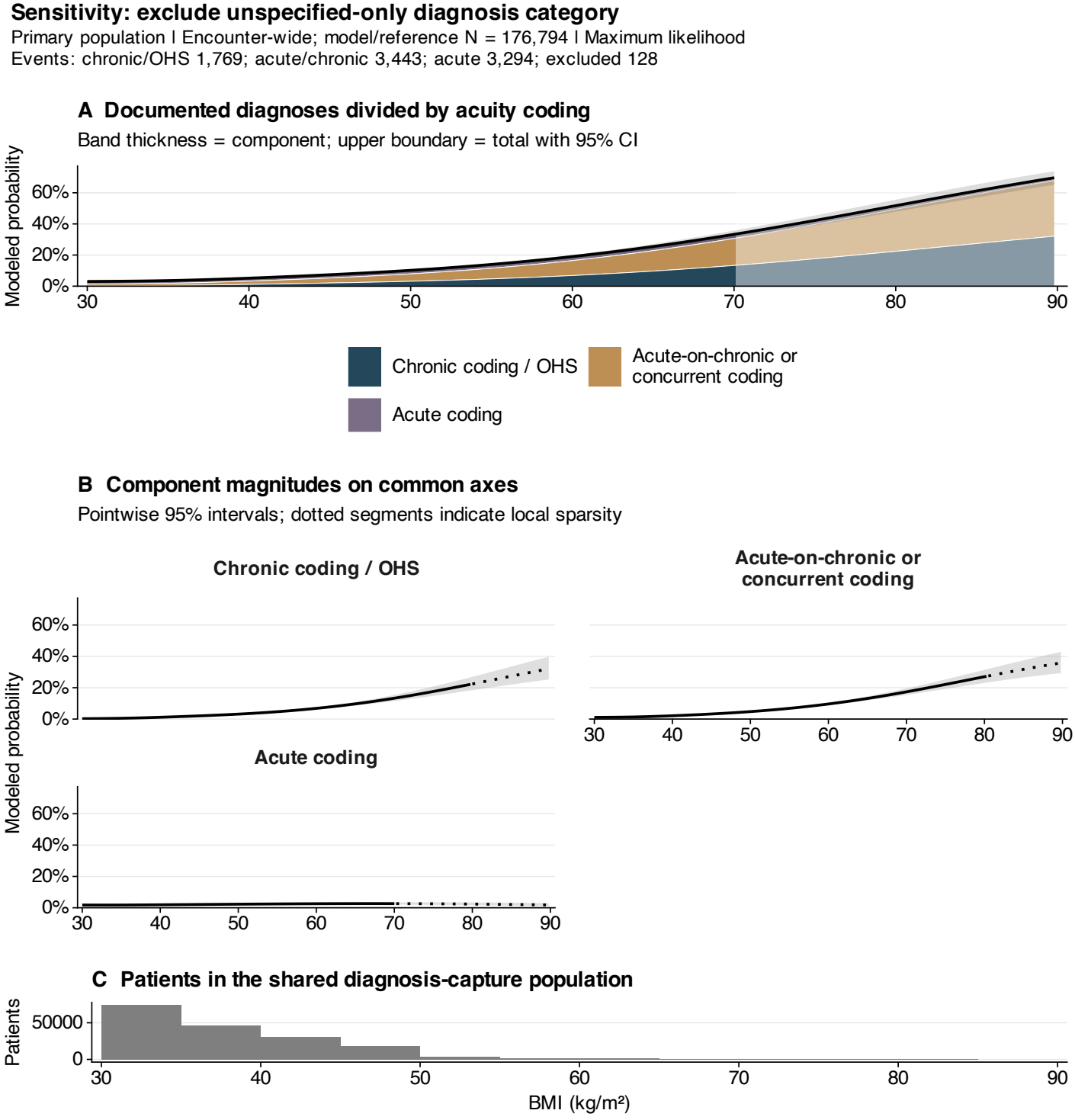



Figure 12: Unpenalized four-category sensitivity excluding unspecified-only encounters and standardizing over the remaining patients. Encounter-wide coding, no gas requirement; inherited adjustment and primary spline basis. Bands partition the composite; thickness gives component probability. Shading: pointwise 95% intervals from joint information covariance, conditional on the empirical reference and basis. Fading/dots: local sparsity; histogram N equals model/reference N. Acute-on-chronic includes explicit or concurrent acute/chronic coding. OHS (obesity hypoventilation syndrome) is analytically chronic; this does not establish chronic respiratory failure. Source selection and diagnostic incorporation limit physiological interpretation.

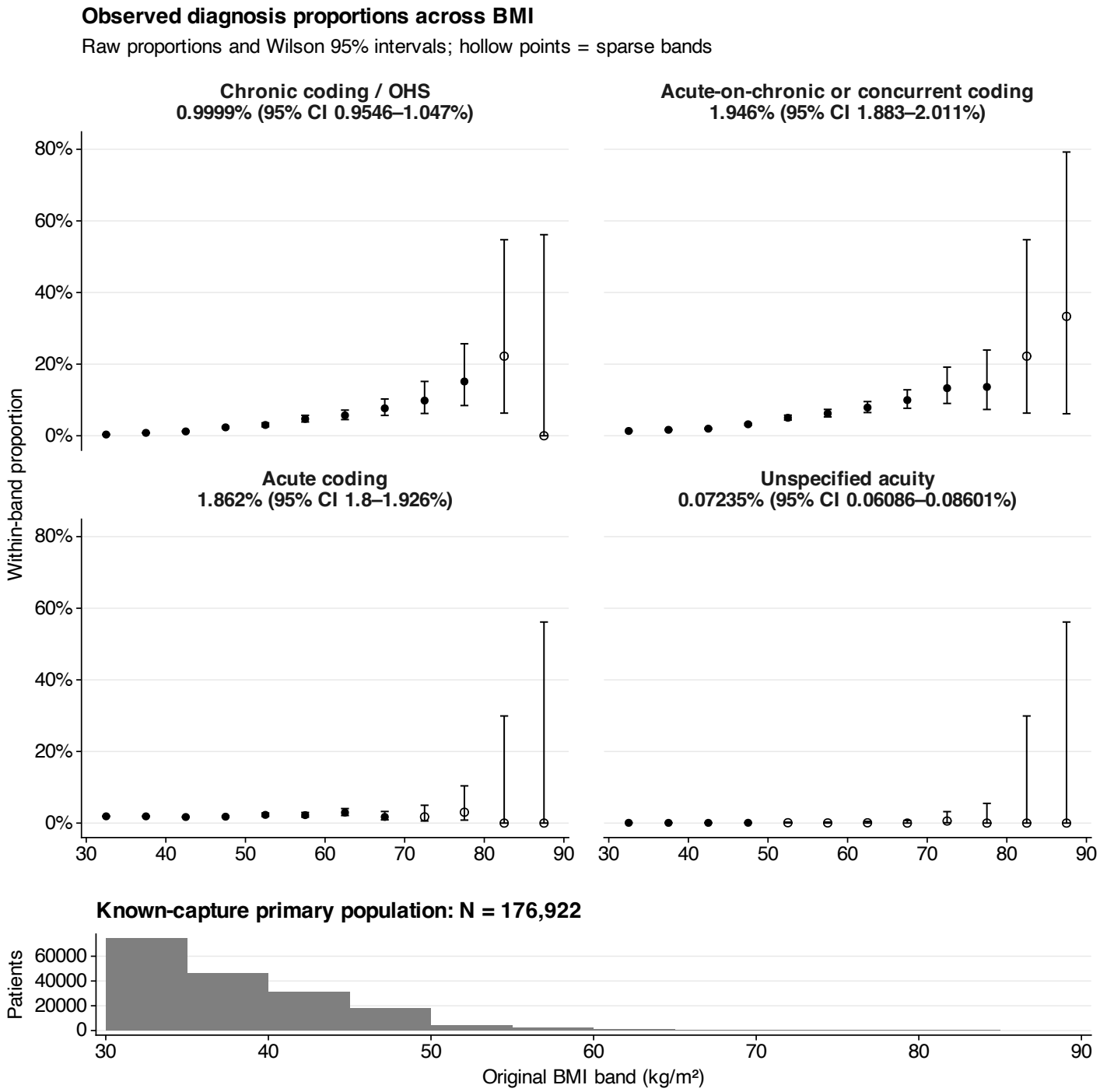



Figure 13: Observed encounter-wide diagnosis proportions within five-unit BMI bands of the covariate-complete primary population with known capture; no gas requirement. These are unadjusted descriptions, not fitted curves. Error bars are Wilson 95% intervals; hollow points mark sparse patient/event/non-event counts. The histogram shows each band denominator, shared by every category. Empty bands have no estimate. OHS is grouped with chronic coding analytically; codes do not establish physiology. Overall percentages and their Wilson intervals are printed in each panel; no-target coding is the complement of the four displayed categories.

Table 45: Original composite and subtype reconciliation

| metric | n | status | analysis population id |
|---|---|---|---|
| complete capture | 176922 | qualified | primary |
| incomplete capture | 0 | qualified | primary |
| unavailable capture | 0 | qualified | primary |
| original composite positive | 8634 | qualified | primary |
| classified positive | 8634 | qualified | primary |

Diagnostic codes reflect documentation and historical source-entry selection. Acute coding is not acidemia, and chronic coding is not evidence of physiological compensation. OHS includes obesity in its definition; a BMI association in that component may partly reflect diagnostic incorporation. The audit retains the five-code patterns, explicit versus concurrent coding, and OHS versus chronic respiratory-failure composition by BMI and source-entry subgroup. This supplement does not establish incident disease or causal effects.

## 4 Discussion and example conclusion

The primary analysis estimates BMI relationships among non-OBESITY-supported indexes while preserving the same clinical episode. Excluding an obesity-only entry route does not remove outcome-dependent entry or selective gas measurement from the retained population. No weighting or transport to excluded indexes is attempted: near-absent gas observation in the excluded entry route does not supply an independent physiological comparator. The overlapping supporting analyses remain in the audit and do not constitute between-group tests.

A defensible conclusion should describe the direction and shape actually supported by the primary curves and the outcome-dependent selection exposed by the evidence audit. No claim of preventable disease or treatment benefit follows from the secondary scenario. Selection into the original files, selective gas measurement, BMI timing and measurement uncertainty, residual confounding and sparse tails remain limitations.

The accompanying audit retains full model output, reconciliation and supporting analyses. Source acceptance, numerical review, visual inspection and investigator scientific approval are distinct states.

**Example wording for the revised primary population:**

> **Example conclusion, pending investigator review:** We estimated BMI–hypercapnia probability relationships within the fixed-index, non-OBESITY-supported source-entry population. Outcome-specific contrasts and their uncertainty are reported separately; they do not constitute a between-group test. Gas-based estimates describe patients with usable blood-gas evidence in this selected acute-care cohort; diagnosis-based estimates describe patients with known diagnosis capture. The primary association remains subject to outcome-dependent entry and selective gas measurement. Secondary one-unit BMI-shift contrasts are descriptive model scenarios and do not establish effects of weight loss or cases prevented.